

F515 汽车手动变速器 维修手册

MAINTAIN MANUAL OF THE F515 GEAR BOX

DYMOS

北京岱摩斯变速器有限公司

版本及更改说明

NO.	版本	变更者	日期	变更事项

前言

欢迎您使用我公司 F515 汽车变速器！

- 1、本维修手册适用于北京岱摩斯变速器有限公司 F515 变速器以及维修。
- 2、本维修手册表达了该变速器各零部件、各总成之间的联系，主要目的是给用户以及维修人员在维护和维修 F515 变速箱时提供一个参考。
- 3、我公司本着“精益求精，用户至上”的原则，竭诚为广大用户服务。诚望广大用户在使用过程中提供宝贵意见。

因变速器故障给您造成的不便，在此谨表示深深的歉意！

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北京岱摩斯变速器有限公司

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一、概要

1、F515 变速器

F515 汽车变速器为手动机械变速器：其前进档均采用惯性同步器换档机构，具有换档平顺、轻快，寿命长等优点；各档齿轮高重合度设计，保证良好的噪音水平。

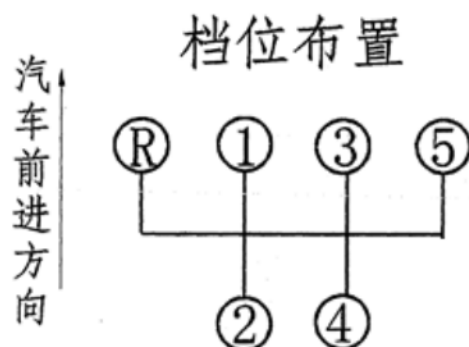
该变速器的操纵机构采用换挡塔式设计，结构简化，零件减少，质量轻；选、换档旷量小，换档灵活、准确，手感更清晰。采用滑移拨叉设计，滑动轴承配合，效率更高，换档更为顺畅。倒档还设有预同步机构，增强了驾驶舒适性和可靠性。

该变速器的结构紧凑，承载能力大，压铸铝合金外壳，产品轻巧，外形美观。

2、技术参数

F515 变速器基本参数		
变速器型号		F515
匹配发动机		473QE
最大输入扭矩 (N.m)		145 N.m
各档速比	1	3.615
	2	1.962
	3	1.286
	4	1.036
	5	0.839
	Rev	3.545
	Final	4.059
中心距 (mm)		66
润滑油		75W-90
润滑油容量 (L)		1.9L±0.05
质量 (Kg)		35.3

档位示意图：



3、螺栓拧紧力矩一览表

使用部位	规格	数	预涂胶	拧紧力矩(N.m)	备注
变速箱壳体 x 离合器壳体	M8×1.25	17		19.6~26.5	
后盖 x 变速箱壳体	M8×1.25	8		19.6~26.5	
空档开关 x T/M 壳体	M18×1.5	1	√	39.2~58.9	
注油塞 x T/M 壳体	M16×1.5	1		58.9~78.5	
放油塞 x T/M 壳体	M16×1.5	1		58.9~78.5	
轴承挡圈 x T/M 壳体	M8×1.25	3	√	14.7~21.6	
差速齿轮 x 差速器壳	M12×1.25	8	√	127.5~137.3	
惰轮齿轮轴 x T/M 壳体	M10×1.25	1		42.2~54.0	
输入轴锁紧螺母	M22×1.5	1		137.3~157.0	需要 Caulking
换挡杆 x T/M 壳体	M8×1.25	2		19.6~26.5	
换挡总成 x T/M 壳体	M8×1.25	4	√	19.6~26.5	
倒挡开关 x T/M 壳体	M14×2.0	1	√	39.2~58.9	
吊钩 x T/M 壳体	M10×1.25	1		39.2~49.1	
拉索支架 x T/M 壳体	M8×1.25	3		14.7~21.6	
感应螺栓	M14×1.5	3	√	39.2~58.9	
拨叉固定螺栓	M12×1.75	1		54.0~58.9	

二、故障诊断与检查

1、常见故障及排除方法

故障症状	可疑部位	排除方法
异常噪音	油位（低）	加油至规定位置
	油（错误）	更换规定使用油品
	离合器分离轴承（磨损或损坏）	检查、更换
	离合器盘（磨损或损坏）	检查、更换
	同步器齿环（输入轴）（磨损或损坏）	更换
	同步器齿环（输出轴）（磨损或损坏）	更换
	齿轮（输入轴）（磨损或损坏）	更换
	齿轮（输出轴）（磨损或损坏）	更换
	轴承（输入轴）（磨损或损坏）	更换
	轴承（输出轴）（磨损或损坏）	更换
	轴承（差速器）（磨损或损坏）	更换
漏油	油位（高）	使油面至规定位置
	垫密圈（损坏）	更换
	油封（磨损或损坏）	更换
换档困难或不能换档	油品型号（错误）	更换规定使用油品
	离合器（卡滞）	检查、调整
	选换挡轴分总成（磨损或损坏）	检查、更换
	控制拉锁（故障）（选档）	检查、调整
	控制拉锁（故障）（换档）	检查、调整
	换档拨叉（输入轴）（磨损）	更换
	换档拨叉（输出轴）（磨损）	更换
	同步器齿环（输入轴）（磨损或损坏）	更换
	同步器齿环（输出轴）（磨损或损坏）	更换
	同步器滑块（输入轴）（磨损或损坏）	更换同步器滑块

	同步器滑块（输出轴）（磨损或损坏）	更换同步器滑块
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(续)

故障症状	可疑部位	排除方法
换档困难或不能换档	齿轮（输入轴）（磨损或损坏）	更换
	齿轮（输出轴）（磨损或损坏）	更换
	同步器齿套（输入轴）（磨损或损坏）	更换
	同步器齿套（输出轴）（磨损或损坏）	更换
跳档或换挡杆移动过度	换档拨叉（输入轴）（磨损）	更换
	换档拨叉（输出轴）（磨损）	更换
	同步器齿套（输入轴）（磨损或损坏）	更换
	同步器齿套（输出轴）（磨损或损坏）	更换
	齿轮（输入轴）（磨损或损坏）	更换
	齿轮（输出轴）（磨损或损坏）	更换
	轴承（输入轴）（磨损或损坏）	更换
	轴承（输出轴）（磨损或损坏）	更换

三、变速器润滑油检查与更换

1、检查变速器润滑油

- 1) 将车辆停在水平地面上。
- 2) 拆下变速器注油螺塞和垫密封圈。
- 3) 检查并确认加油量为 $1.9 \pm 0.05\text{L}$ 。
- 4) 安装新垫密封圈和注油螺塞。

拧紧力矩： $49.1 \pm 9.8\text{N.m}$

2、润滑油更换

2.1、排空润滑油

- 1) 将车辆停放在水平地面上。
- 2) 拆下变速器注油螺塞和垫密封圈。
- 3) 使用套筒扳手拆下放油螺塞和垫密封圈以排空润滑油。
- 4) 使用套筒扳手安装新垫密封圈和放油螺塞。

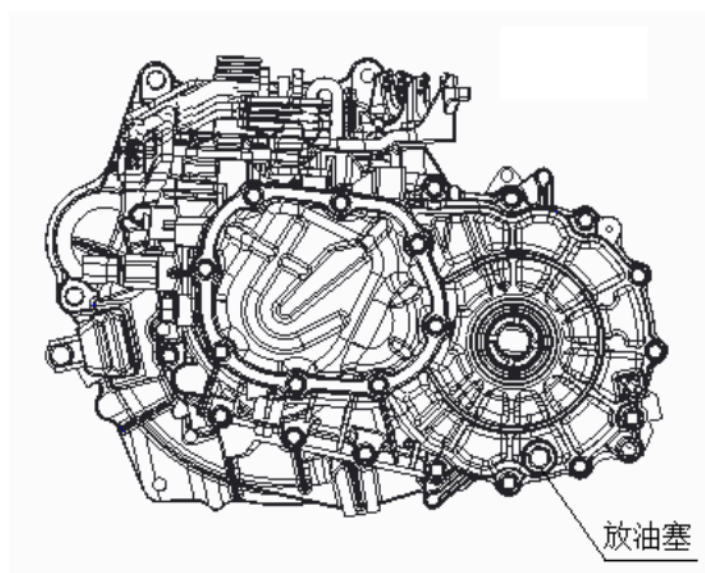
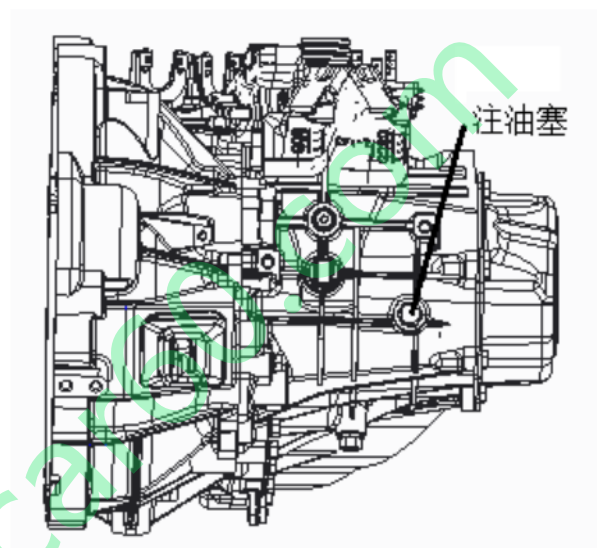
拧紧力矩： $49.1 \pm 9.8\text{N.m}$

2.2、加注润滑油

- 1) 加注润滑油。
- 润滑油：道达尔 75W/90
- 标准容量： $1.9 \pm 0.05\text{L}$

- 2) 安装新垫密封圈和注油螺塞。

拧紧力矩： $49.1 \pm 9.8\text{N.m}$



车辆保持水平状态。

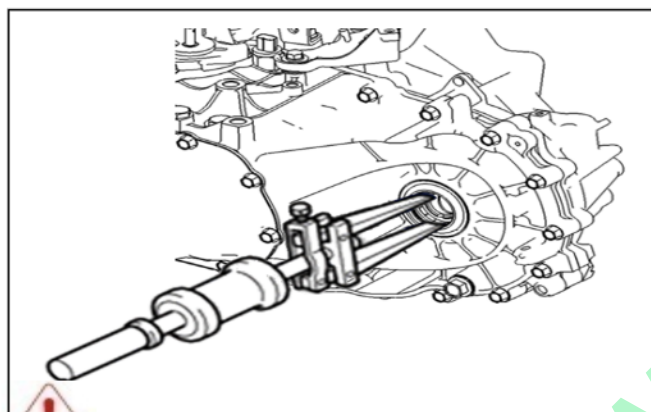
- 变速器润滑油过多或过少均可能导致故障

四、差速器油封的更换

1、差速器油封的更换

拆卸

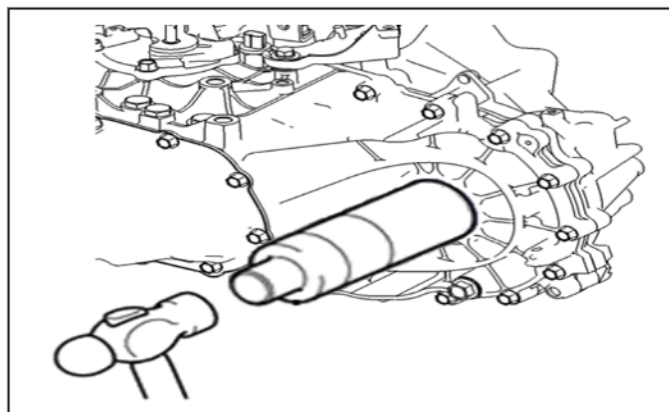
- 1) 使用工装，从变速器总成上拆下左半轴油封。
- 2) 使用工装，从离合器总成上拆下右半轴油封。



- 不要损坏左半轴油封唇口。
- 加注润滑油时，确保车辆保持水平状态。
- 变速器润滑油过多或过少均可能导致故障。

安装

- 1) 在新的左半轴油封唇口上涂抹通用润滑脂。
- 2) 使用工装和锤子，将左半轴油封安装到变速器总成上，压装至壳限位面位置。
- 3) 在新的右半轴油封唇口上涂抹通用润滑脂。
- 4) 使用工装和锤子，将右半轴油封安装到变速器总成上，压装至壳限位面位置。



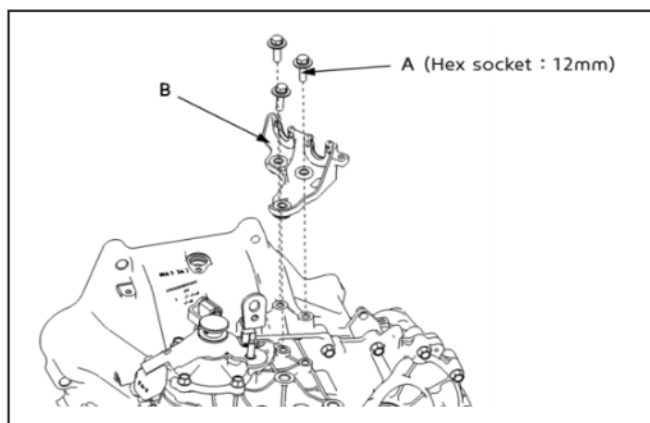


- 不要损坏右半轴油封唇口。

拆卸

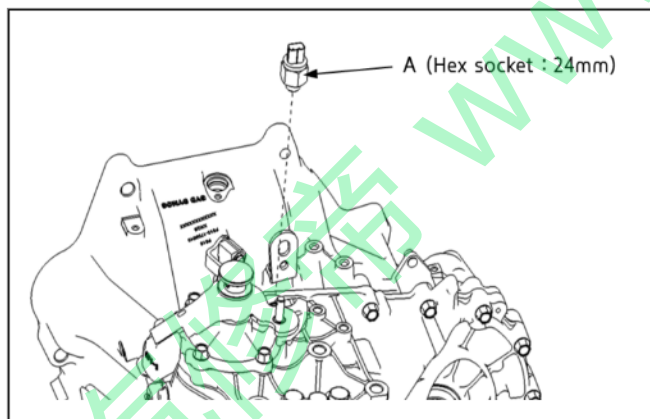
- 1) Remove 3-seal bolts (A) and cable bracket (B).

移除 3 个密封螺栓(A)和拉索支架(B)。



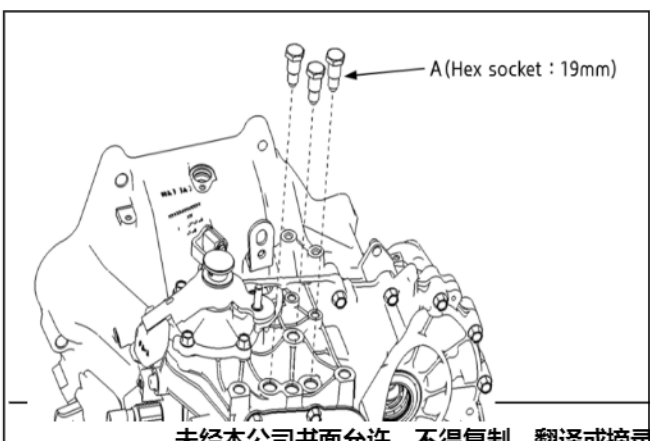
- 2) Remove back up lamp switch (A).

移除倒档开关(A)



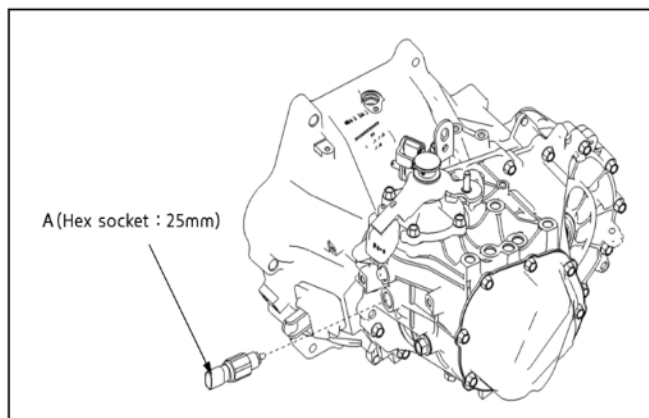
- 3) Remove 3-detent pin (A).

移除 3 个定位销(A)



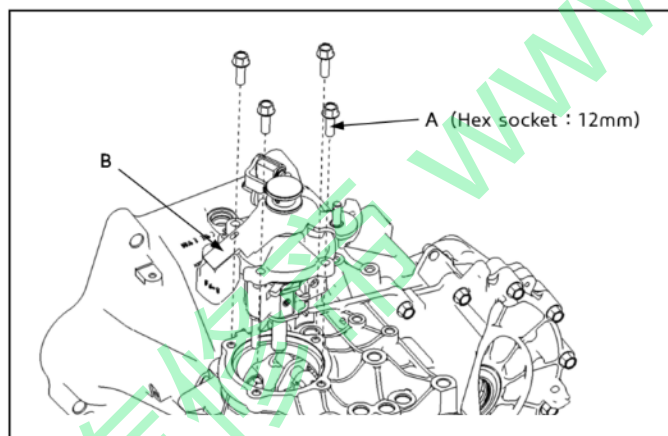
4) Remove neutral switch (A).

移除空挡开关(A)



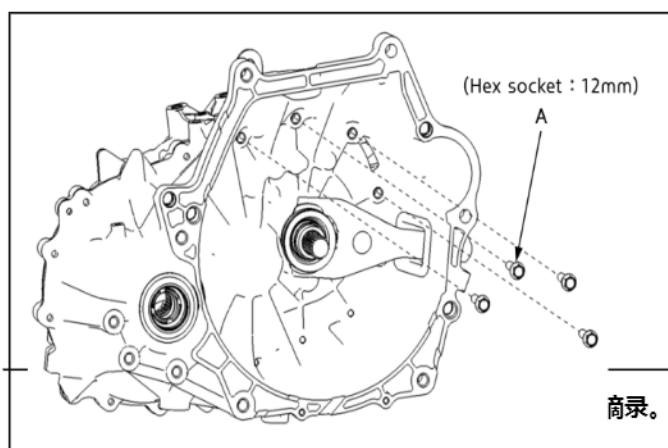
5) Remove 4-seal bolts (A) and control shaft complete (B).

移除 4 个密封螺栓(A)和换挡总成(B)



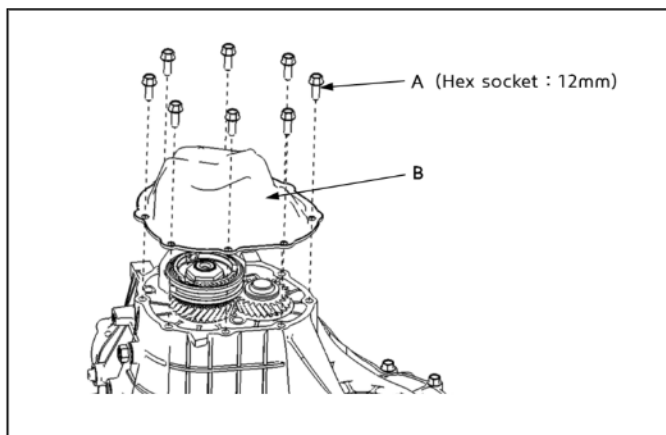
6) Remove 4-flange bolts (A).

移除 4 个法兰螺栓(A)



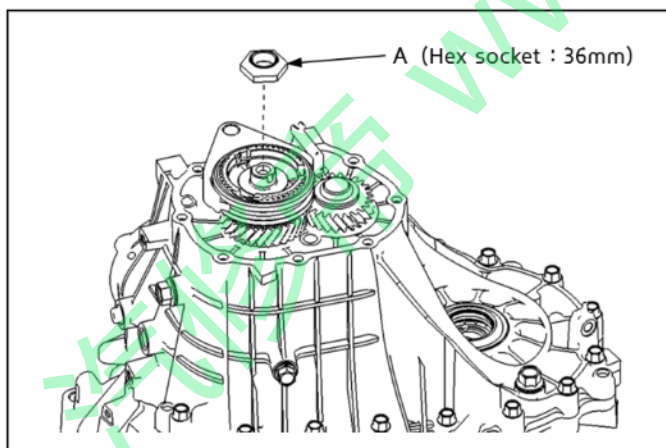
7) Remove 8-flange bolts (A) and rear cover (B).

拆卸 8 个法兰螺栓(A)和后盖(B)



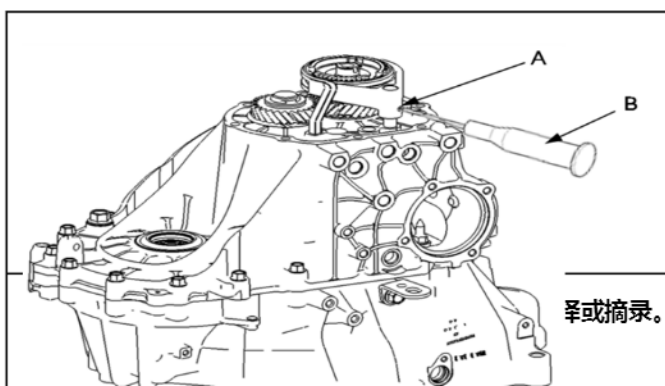
8) Remove locking nut (A).

拆下锁紧螺母(A)



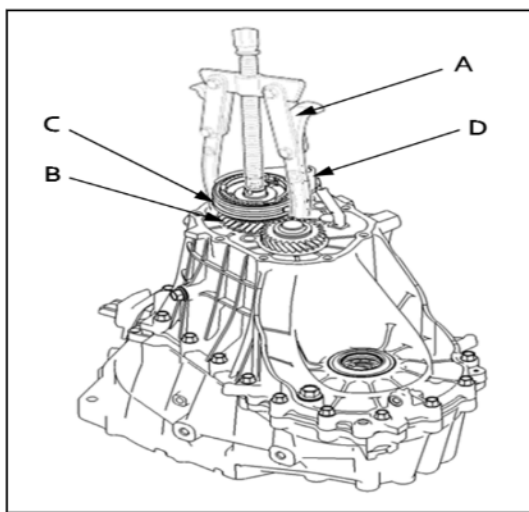
9) Remove spring pin (A) by using a chisel (B).

用锤子(B)拆卸弹簧销(A)



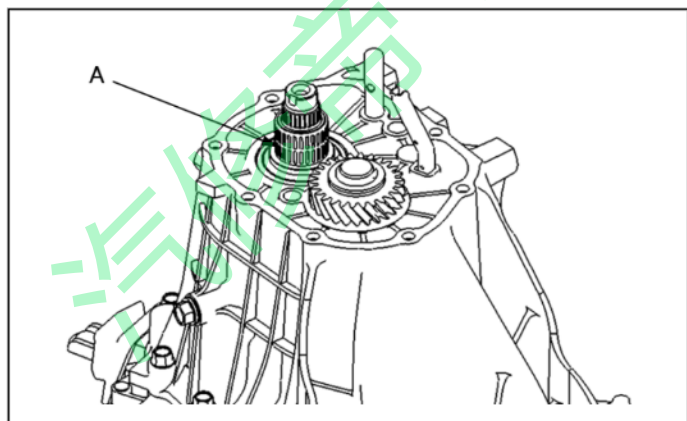
10) After installing a gear puller (A) to 5th gear (B), disassemble 5th hub & sleeve(C) - 5th shift fork (D) at once.

在将齿轮拆卸器(A)置于五档齿轮(B)处，拆卸 5 档齿毂齿套(C) - 5 档拨叉(D)。



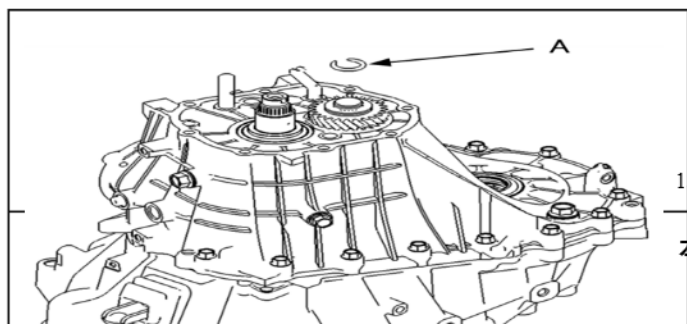
11) Remove the needle bearing (A).

拆下滚针轴承(A)



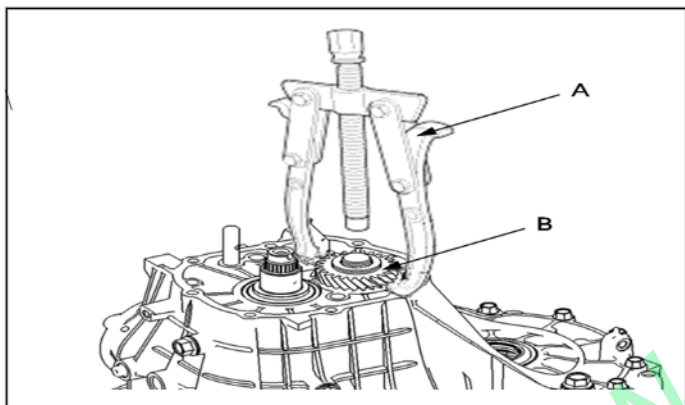
12) Remove the snap ring (A) by using a snap ring plier.

使用卡簧钳拆卸卡环(A)。



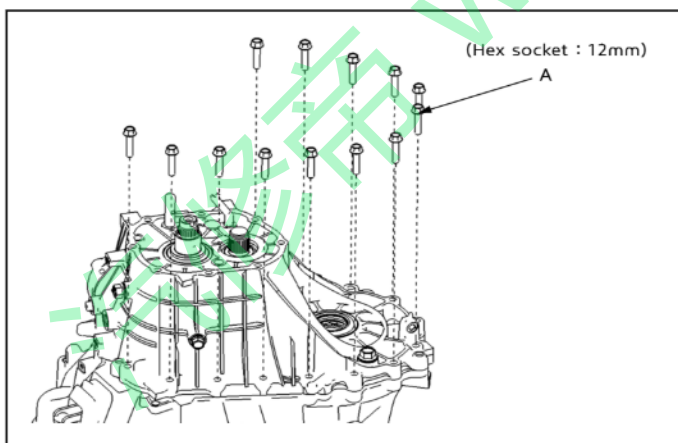
13) Remove the 5th output gear (A) by using a gear puller (B).

使用齿轮拆卸器(B)拆卸 5 档输出齿轮(A)



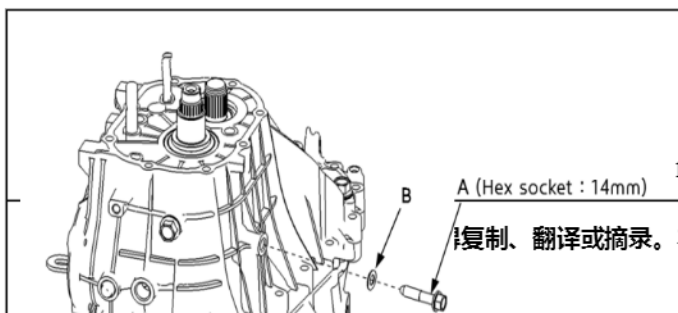
14) Remove 13-flange bolts (A).

拆卸 13 个法兰螺栓(A)



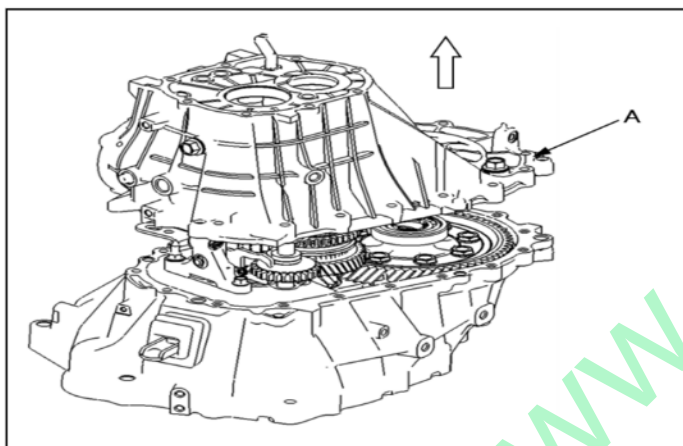
15) Remove reverse idler bolt (A) and gasket (B).

拆卸倒档惰轮螺栓(A)和垫圈(B)。



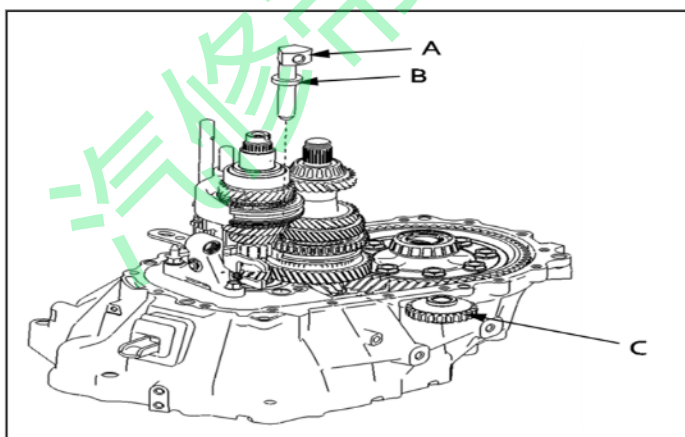
16) Pull the transmission case (A) upward.

将变速箱(A)向上拉动



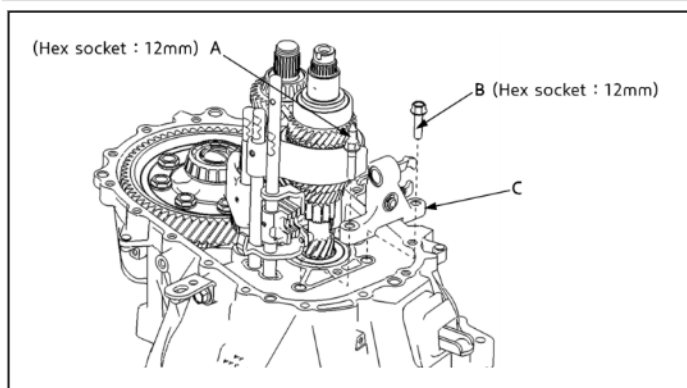
17) Remove reverse idler shaft (A), reverse spacer (B) and reverse idler gear (C).

拆卸倒档杆(A)、倒档间隔器(B)和倒档惰轮(C)。



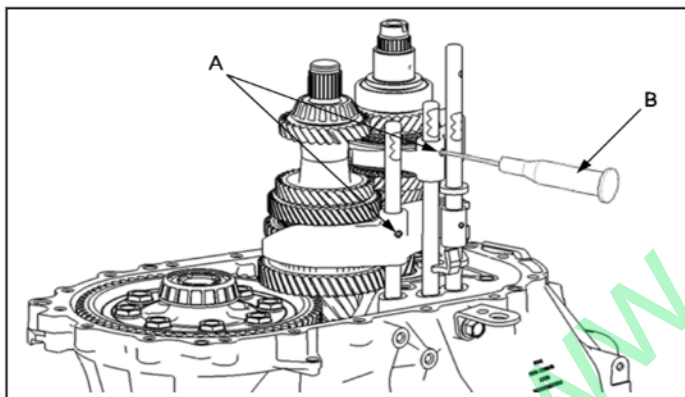
18) Remove inter lock bolt (A), flange bolt (B) and reverse shift lever (C).

拆去联锁螺栓(A)、法兰螺栓(B)和倒档换挡杆(C)。



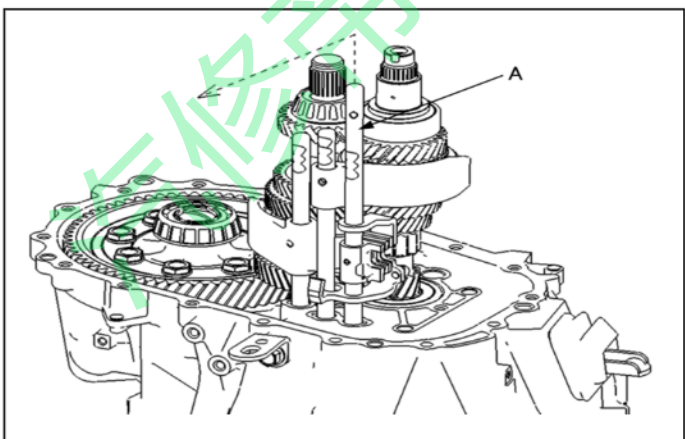
19) Remove 2-spring pins (A) by using a chisel (B).

用凿子(B)移除 2 个弹簧销(A)



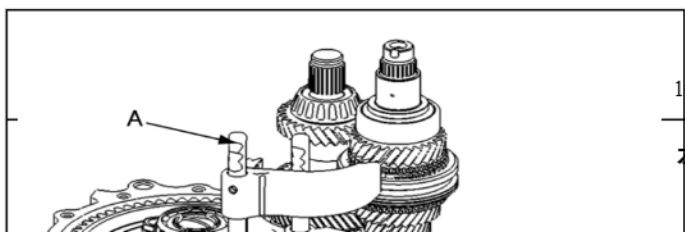
20) Remove 5&R shift rail sub assembly (A) after pulling upward 5&R shift rail.

在向上拉动 5&R 换档组件后，拆下 5&R 换档组件(A)。



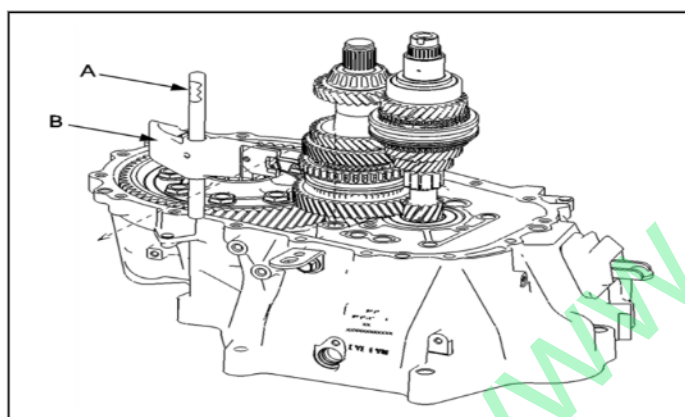
21) Remove 3&4 shift rail sub assembly (A) after pulling upward 3&4 shift rail.

在向上拉动 3&4 换档组件后，移走 3&4 档换档组件(A)



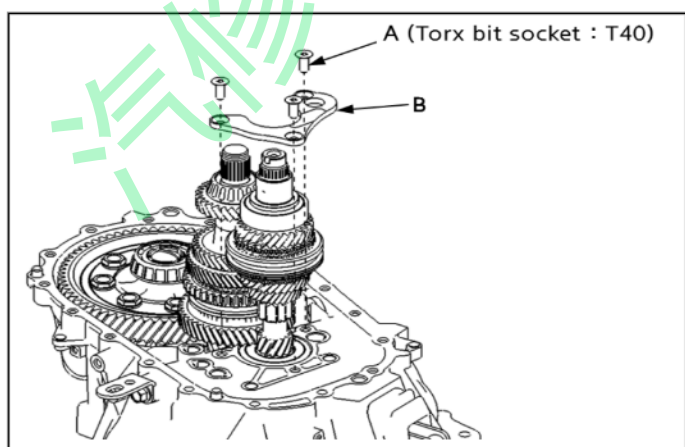
22) Remove 1&2 shift rail (A) and 1&2 shift fork (B) after pulling upward 1&2 shift rail.

拔出 1&2 换档拨叉后，向上拉动 1&2 换档杆(A)和拨叉(B)



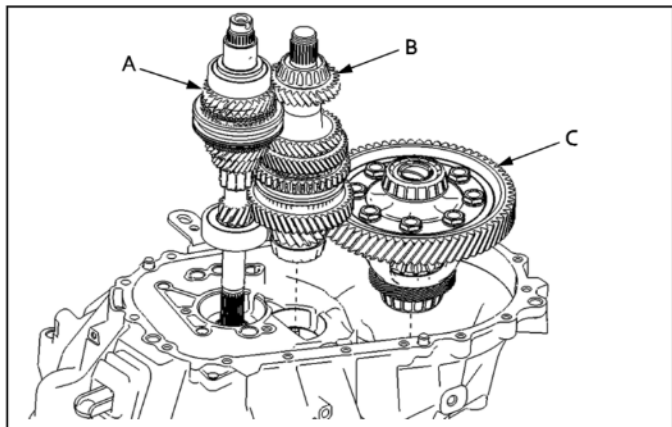
23) Remove 3-seal bolts (A) and bearing retainer (B).

拆卸 3 个密封螺栓(A)和轴承保持器(B)。



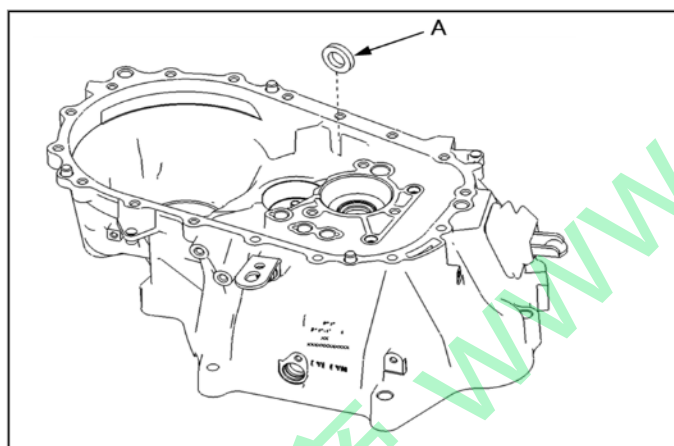
24) Pull upward input shaft (A), output shaft (B) and differential assembly(C) at the same time.

同时向上拉动输入轴(A)、输出轴(B)和差速器总成(C)。



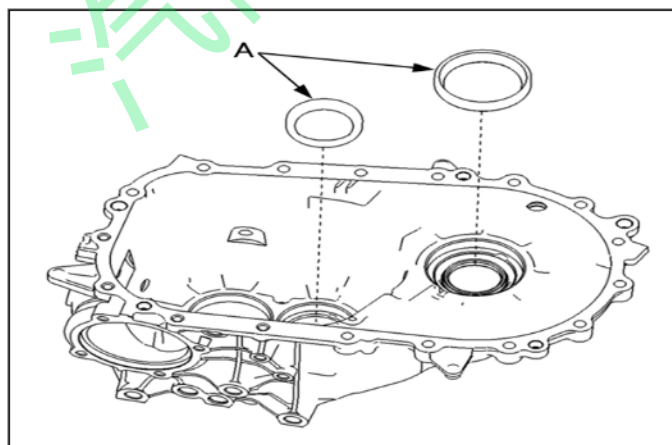
25) Remove magnet (A).

移除磁铁(A)



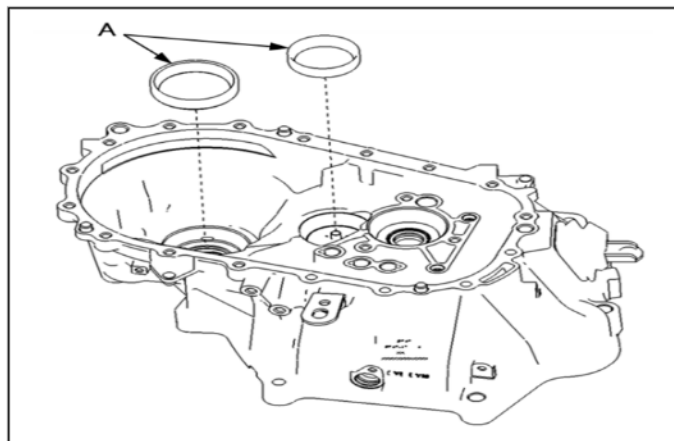
26) Remove the bearing outer races (A) from the clutch housing by using a sliding hammer.

用滑动锤从离合器下壳上拆下轴承套(A)。



27) Remove the bearing outer races (A) from the transmission case by using a sliding hammer.

用滑动锤从变速箱上壳中拆下轴承套 (A)。



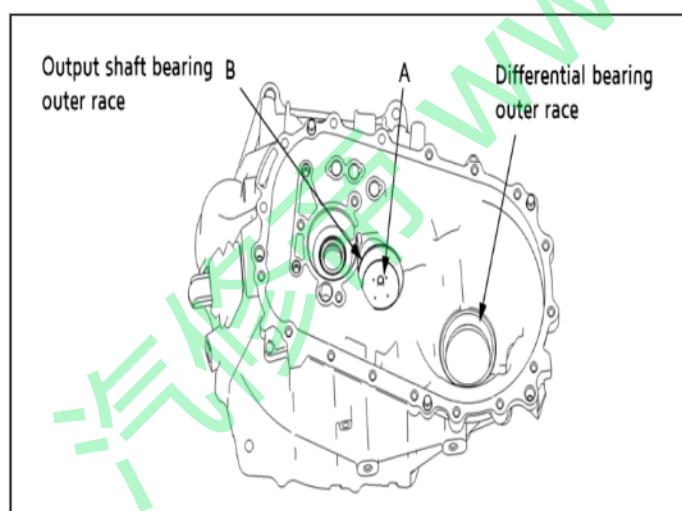
安装

1) Install the bearing outer races into the clutch housing.

将轴承套安装到离合器壳体中。

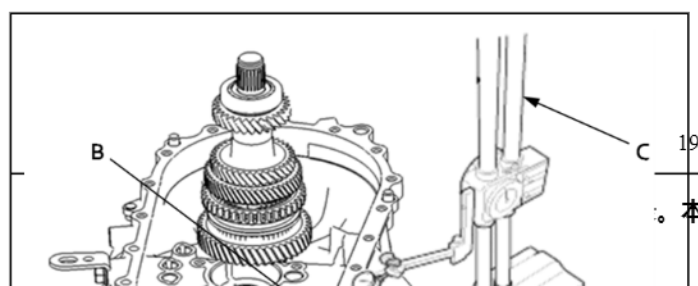
※ Insert oil guide(A) before pressing the output shaft bearing outer race(B).

※ 在压入输出轴轴承套(B)之前插入导油盖(A)。



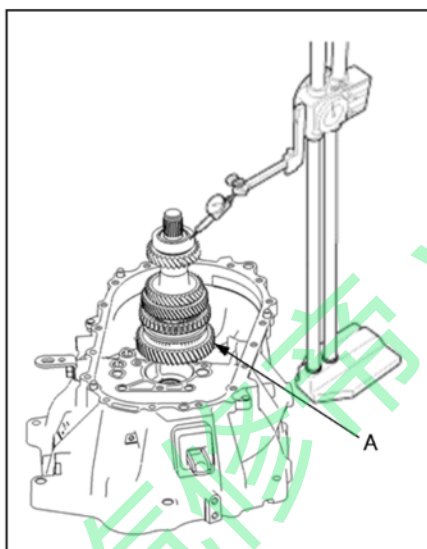
2) Measure the heights of 3 points on the surface of the clutch housing(A) with a height gauge(C) to set the surface(B) as '0' height.

用高度仪(C)测量变速器下壳(A)端面 3 点的高度，将端面(B)设为高度“0”。



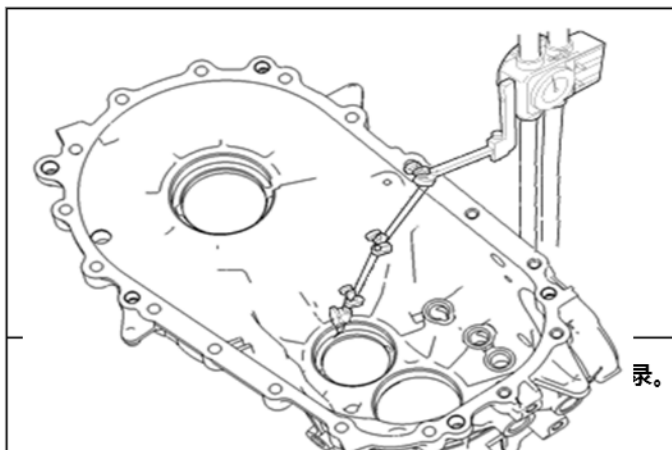
3) Measure the height (Ho) of the output shaft (A) with the bearing outer race assembled. (Preload : 70 kgf).

在组装轴承套的状态下测量输出轴(A)的高度(HO)。(预加载：70 kgf)



4) Measure the height (Ho') of the output shaft's space from the '0' height on the transmission case.

测量输出轴从“0”高度到上壳的高度(HO)



5) Choose a shim for output shaft by using the formula and below chart and install the shim into the transmission case.

- **Output shaft end play** : $0 \sim 0.05T$ mm

- **Shim thickness** : $Ho' - Ho - \text{Output shaft end play}$

ex) $169 - 168.25 - (0 \sim -0.05) = 0.75 \sim 0.80$

通过公式和下图选择输出轴垫片，并将垫片安装在变速箱中。

-输出轴端间隙： $0 \sim 0.05T$ mm

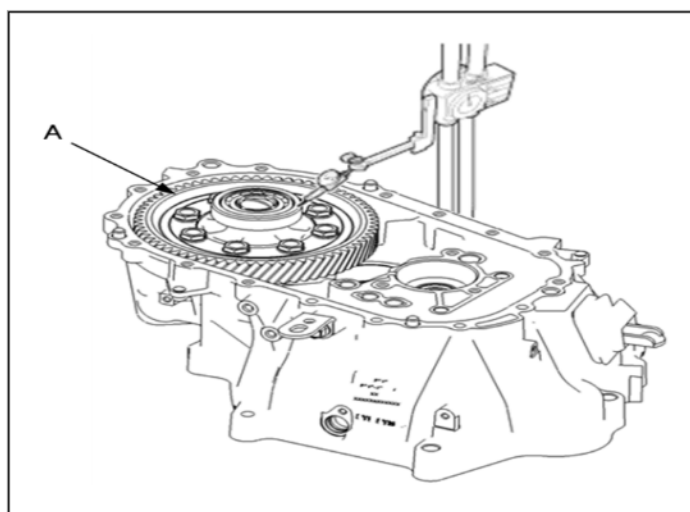
-薄垫片厚度： $Ho' - Ho - \text{输出轴端间隙}$

ex) $169 - 168.25 - (0 \sim -0.05) = 0.75 \sim 0.80$

Part number	Thickness[mm]	I.D number
43203-32052	0.52	52
43203-32055	0.55	55
43203-32058	0.58	58
43203-32061	0.61	61
43203-32064	0.64	64
43203-32067	0.67	67
43203-32070	0.70	70
43203-32073	0.73	73
43203-32076	0.76	76
43203-32079	0.79	79
43203-32082	0.82	82
43203-32085	0.85	85
43203-32088	0.88	88
43203-32091	0.91	91
43203-32094	0.94	94
43203-32097	0.97	97
43203-32100	1.00	00
43203-32103	1.03	03
43203-32106	1.06	06
43203-32109	1.09	09

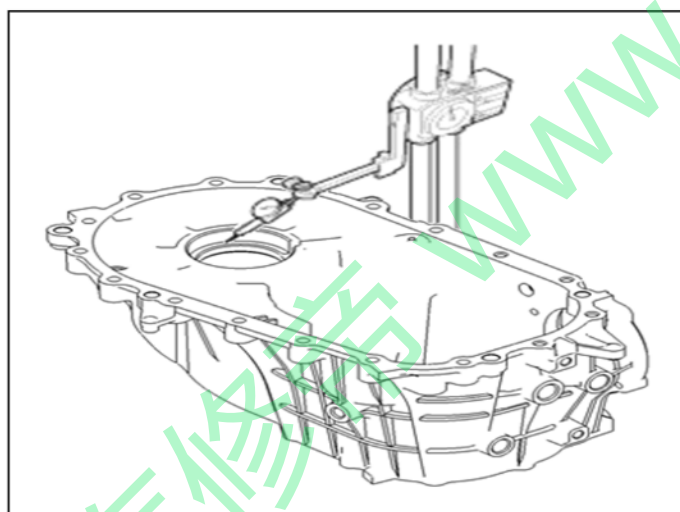
6) Measure the height (Hd) of the differential assembly (A) with the outer bearing race assembled. (Preload: 70 kgf).

测量有轴承套状态下的测量差速器总成(A)的高度(HD)。(Preload: 70 kgf).



7) Measure the height(Hd') of the differential assembly's space from the '0' height on the transmission case.

从“0”位置开始测量差速器总成到上壳的高度(HD)



8) Choose a shim for differential assembly by using the formula and below chart and install the shim into the transmission case.

- **Differential assembly end play** : 0.15T ~ 0.20T mm

- **Shim thickness** : Hd' - Hd – Differential assembly end play

ex) $23.43 - 22.72 - (-0.15 \sim -0.20) = 0.86 \sim 0.91$

使用公式和下图选择差速器总成垫片，并将垫片安装在变速箱中。

-差速器总成端面间隙：0.15t~0.20t

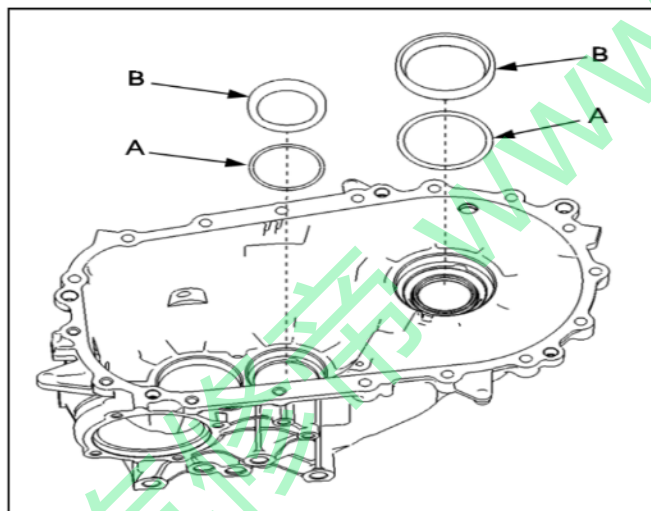
--薄垫片厚度：HD'-HD -差速器装配末端

$$\text{ex) } 23.43 - 22.72 - (-0.15 \sim -0.20) = 0.86 \sim 0.91$$

Part number	Thickness[mm]	I.D number
43331-23071	0.71	71
43331-23074	0.74	74
43331-23077	0.77	77
43331-23080	0.80	80
43331-23083	0.83	83
43331-23086	0.86	86
43331-23089	0.89	89
43331-23092	0.92	92
43331-23095	0.95	95
43331-23098	0.98	98
43331-23101	1.01	01
43331-23104	1.04	04
43331-23107	1.07	07
43331-23110	1.10	10
43331-23113	1.13	13
43331-23116	1.16	16
43331-23119	1.19	19
43331-23122	1.22	22
43331-23125	1.25	25
43331-23128	1.28	28

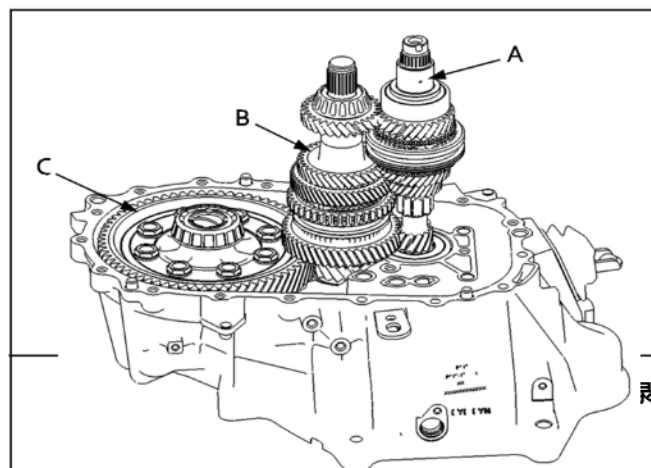
9) Install shims (A) and bearing outer races (B) into the transmission case.

将垫片(A)和轴承套(B)安装到变速箱中。



10) Insert input shaft (A), output shaft (B) and differential assembly (C) into the clutch housing at the same time.

同时将输入轴(A)、输出轴(B)和差速器总成(C)插入离合器壳体中。



11) Install front bearing retainer (A) with 3-seal bolts (B) after spreading thread locker on the screw.

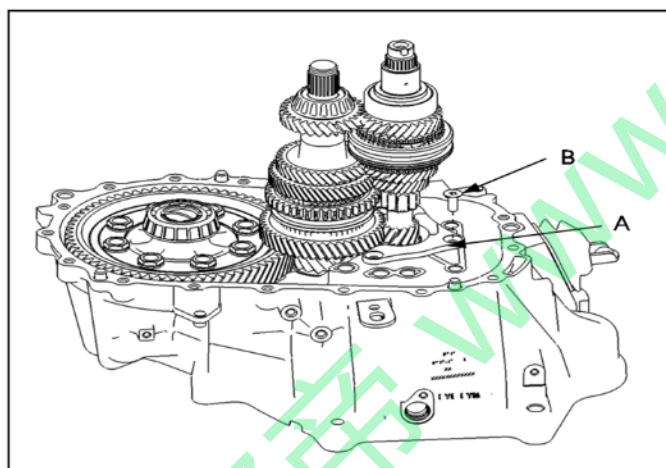
Tightening torque : 14.7 ~ 21.6Nm

(Torx bit socket : T40)

在螺钉上展开螺纹锁紧器后，安装带有 3 个密封螺栓(B)的前轴保持架(A)。

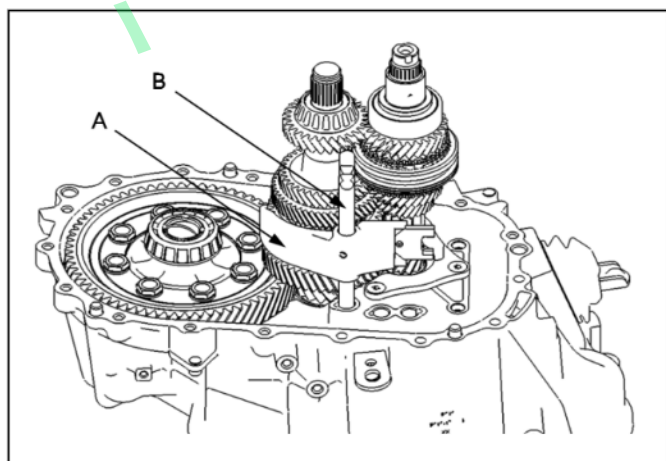
拧紧扭矩：14.7 ~ 21.6nm

(牙轮钻头：T40)



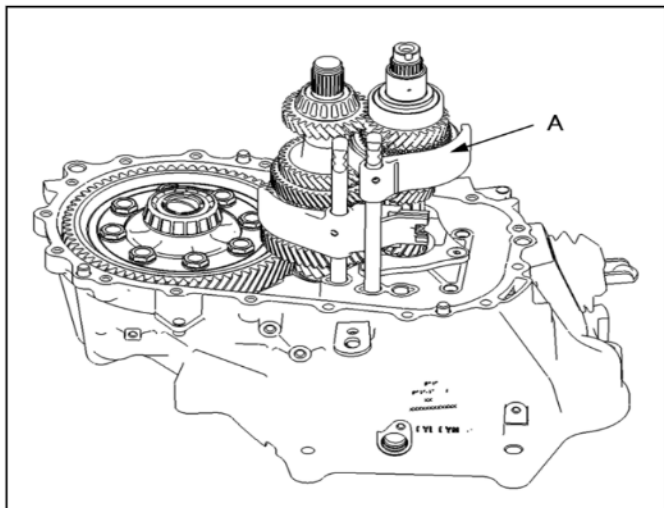
12) Install 1&2 shift fork (A) and 1&2 shift rail(B).

安装 1&2 换档拨叉(A)和 1&2 换档杆(B)。



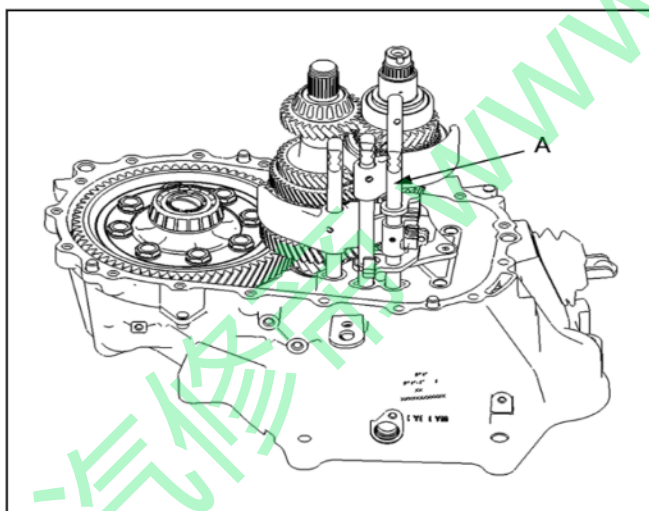
13) Install 3&4 shift rail sub assembly (A).

安装 3&4 换档总成(A)。



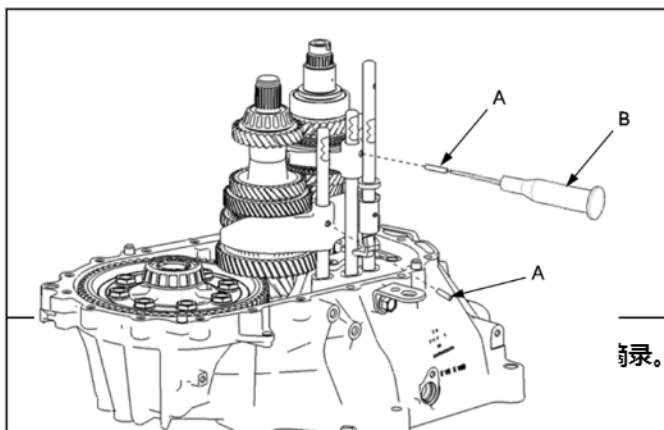
14) Install 5&R shift rail sub assembly (A).

安装 5&R 换档总成(A)。



15) Install spring pins (A) by using a chisel (B).

用凿子(B)安装弹簧销(A)



16) Install reverse shift lever (A) with interlock bolt (B) and flange bolt (C).

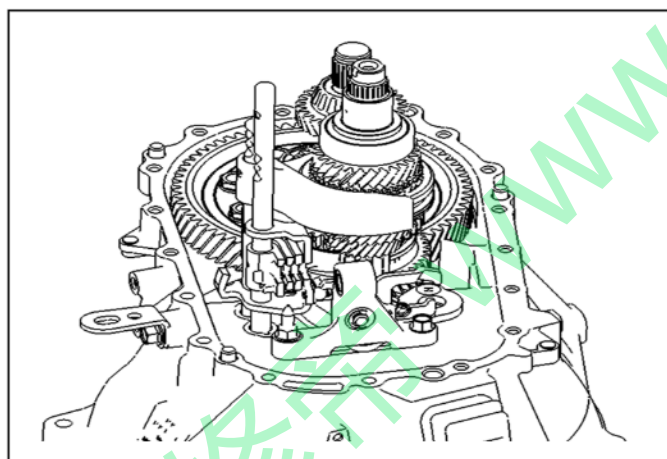
Tightening torque : 19.6 ~ 26.5Nm

(Hex socket : 12mm)

安装带有联锁螺栓(B)和法兰螺栓(C)的倒档杆(A)。

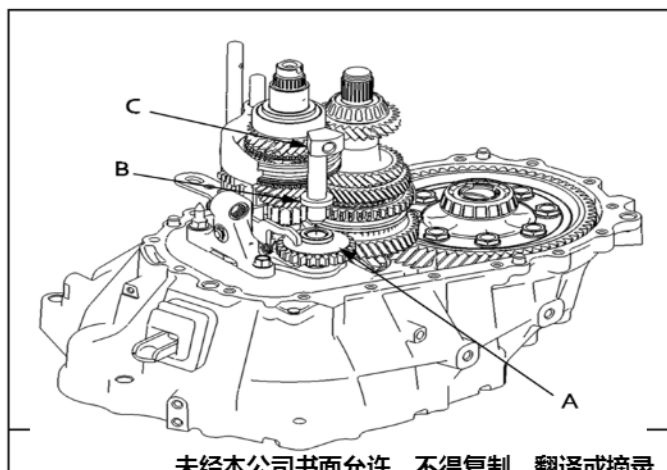
拧紧扭矩 : 19.6 ~ 23.5Nm

(内六角 : 12mm)



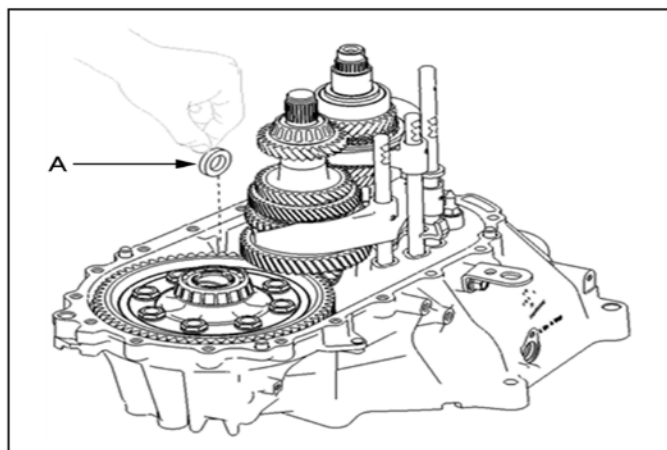
17) Insert reverse idler gear (A), reverse spacer (B) and reverse idler Shaft (C).

插入倒档惰轮(A)，倒档垫片(B)和倒档轴(C)



18) Insert magnet (A) into the groove on the clutch housing

将磁铁(A)插入离合器壳体的凹槽中

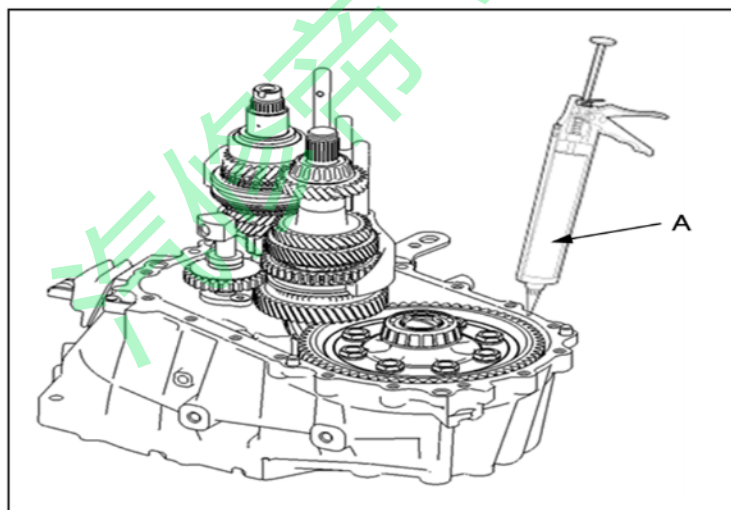


19) Clean the mating surface of clutch housing and spread a fluid gasket (A).

Fluid gasket : Loctite 5060 or equivalent

清洁壳体表面并涂密封胶(A)。

密封胶：乐泰 5060 或类似



20) Install transmission case (A) with 13-flange bolt(B).

※ Check the oil guide(C) before assembling the transmission case.

Tightening torque : 19.6 ~ 26.5Nm

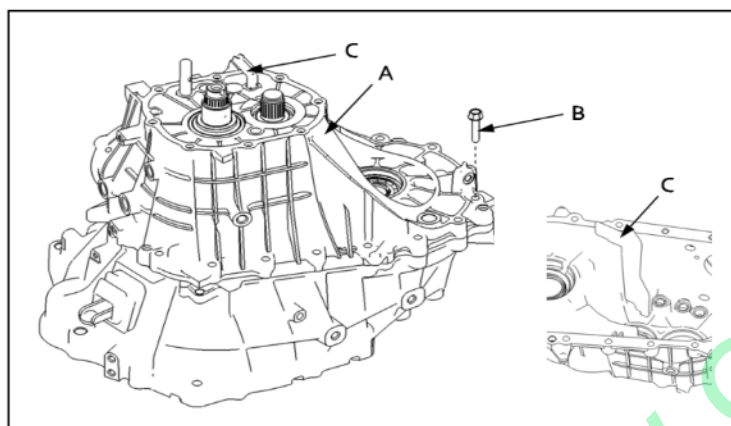
(Hex socket : 12mm)

用 13 个法兰螺栓(B)安装上壳(A)。

※ 在安装上壳前检查导油槽(C)。

拧紧扭矩 : 19.6 ~ 23.5Nm

(内六角 : 12mm)



21) Install gasket (A) and reverse idler bolt (B).

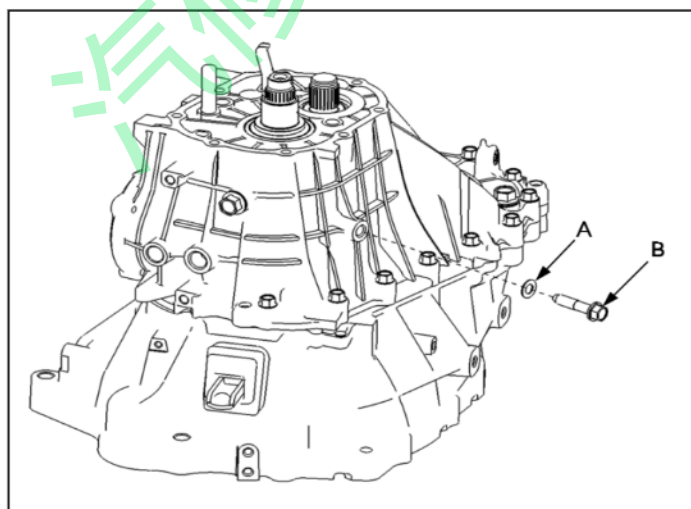
Tightening torque : 42.2 ~ 54.0Nm

(Hex socket : 14mm)

安装垫圈(A)和倒档螺栓(B)。

拧紧扭矩 : 42.2 ~ 54.0Nm

(内六角 : 14mm)



22) Install 4-flange bolt (A).

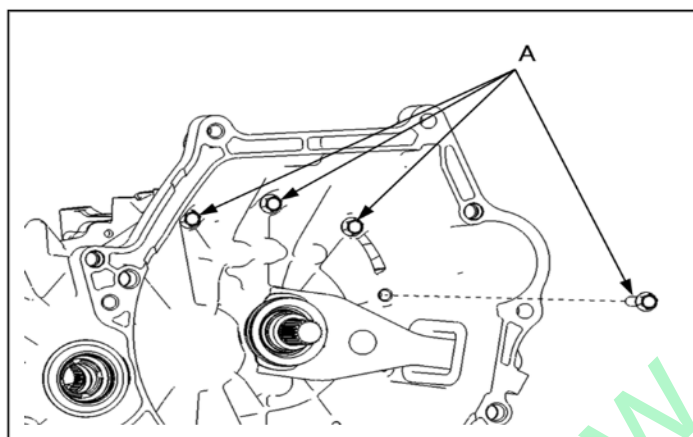
Tightening torque : 19.6 ~ 26.5Nm

(Hex socket : 12mm)

安装 4 个法兰螺栓(A)。

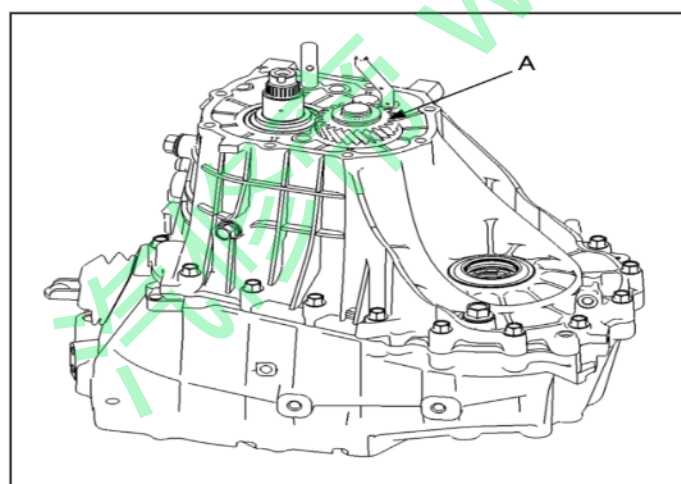
拧紧扭矩 : 19.6 ~ 23.5Nm

(内六角 : 12mm)



23) Press the 5th output gear (A)

压入第 5 档输出齿轮(A)



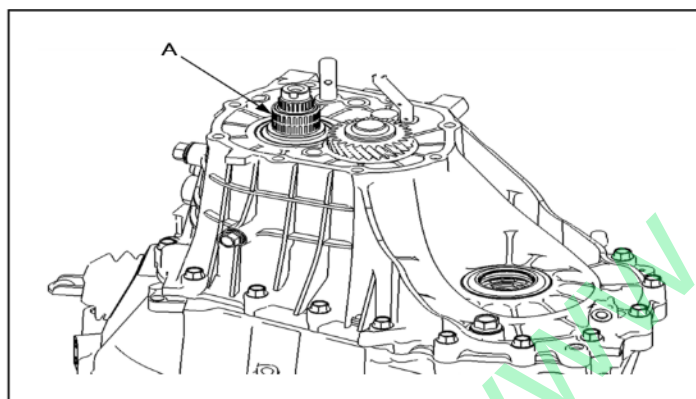
24) Select the snap ring(A) to adjust the gear clearance to 0 ~ 0.05mm and insert snap ring(A) in the slot of output shaft(B) by using snap ring plier.

选择卡圈(A)去调整齿轮间隙在 0~0.05mm 并且用卡簧钳在输出轴(B)槽处插入卡圈(A)

Part number	Thickness[mm]	I.D Marks
43233-32190	1.90	GREEN
43233-32194	1.94	BLUE
43233-32198	1.98	RED
43233-32202	2.02	- (NON)
43233-32206	2.06	BLLACK
43233-32210	2.10	YELLOW
43233-32214	2.14	WHITE
43233-32218	2.18	ORANGE

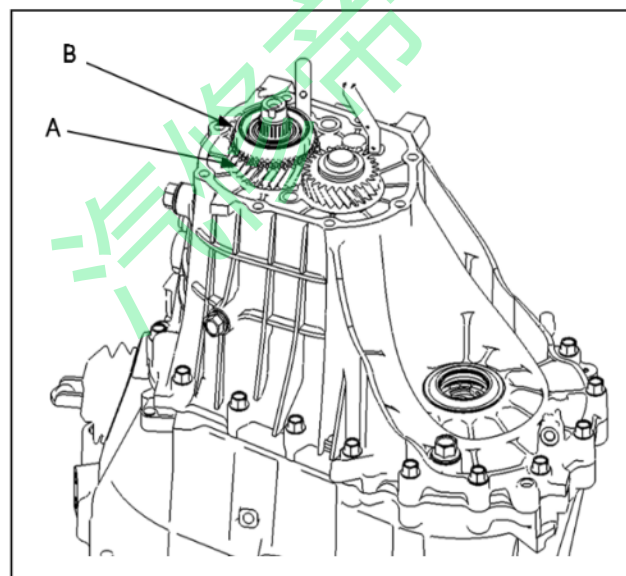
25) Insert the needle bearing (A).

插入滚针轴承 (A)



26) Insert the 5th gear (A) and the synchronizer ring (B).

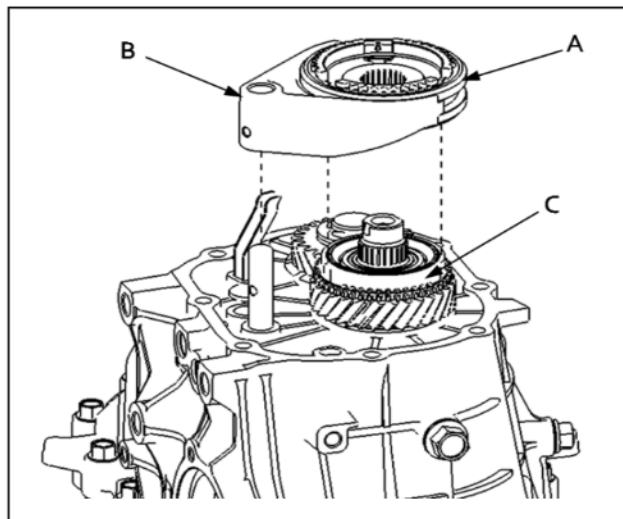
插入五档齿轮(A)和同步器环(B)



27) Press 5th hub assembly (A) and 5th shift fork (B) after aligning boss of synchronizer ring(C) with slot of 5th

hub.

校正同步器环(C)齿套后, 在 5 档齿毂凹槽处压入 5 档齿毂组件(A)和 5 档拨叉(B)



28) Fasten the lock nut (A) and caulk the lock nut (A)

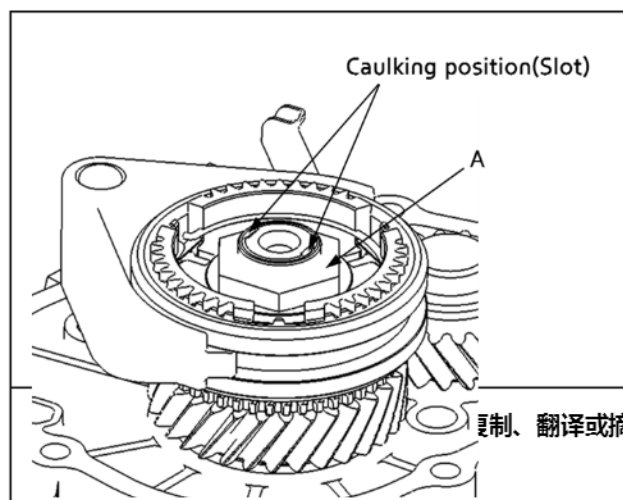
Tightening torque : 137.3 ~ 157.0Nm

(Hex socket : 36mm)

紧固锁紧螺母(A)并锁定锁紧螺母(A)

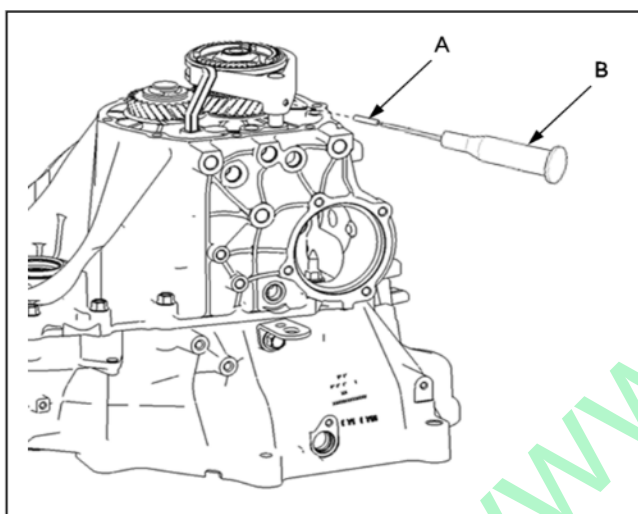
拧紧扭矩 : 137.3 ~ 157.0Nm

(内六角 : 36mm)



29) Install spring pins (A) by using a chisel (B).

用锤子(B)安装弹簧销(A)。

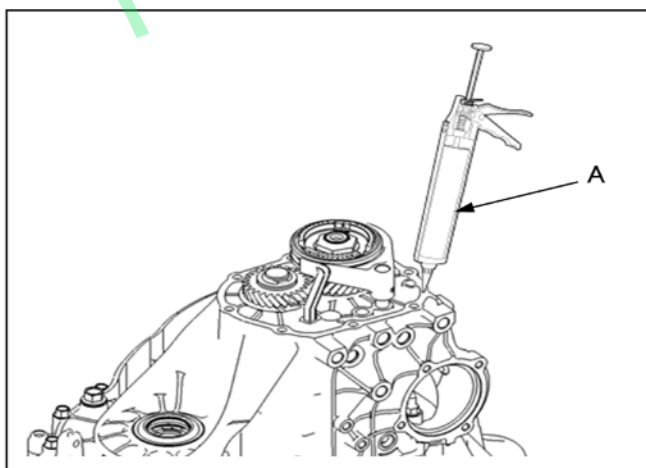


30) Clean the mating surface of transmission case and spread a fluid gasket (A).

Fluid gasket : Loctite 5060 or equivalent

清洗变速器壳啮合面并涂密封胶(A)

密封胶：乐泰 5060 或类似



31) Install rear cover (B) with 8-flange bolt (A).

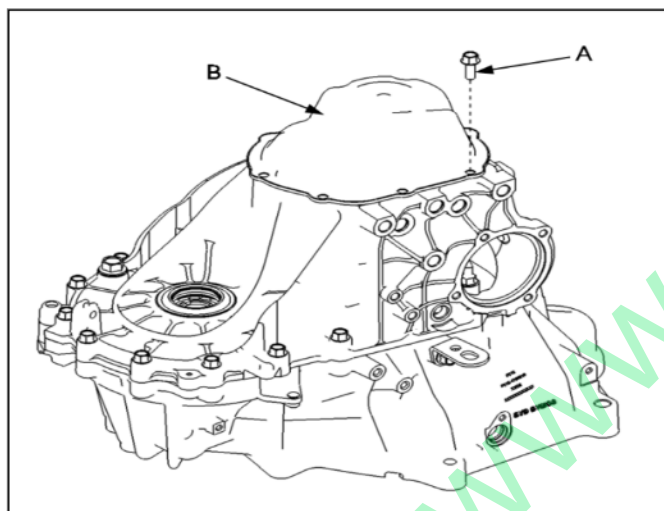
Tightening torque : 19.6 ~ 26.5Nm

(Hex socket : 12mm)

用 8 个法兰螺栓(A)安装上盖(B)

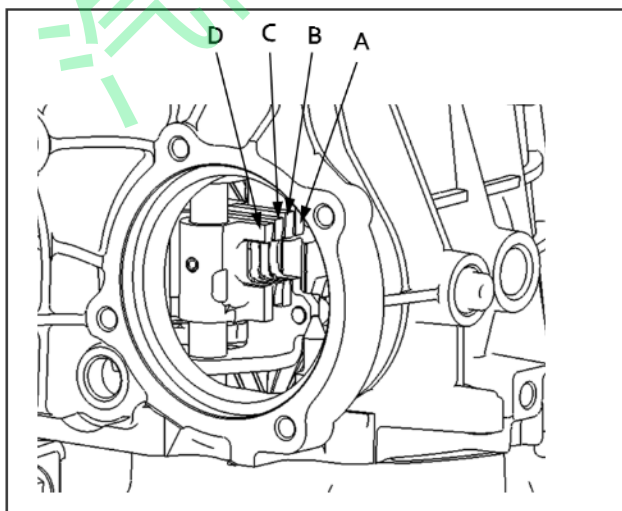
拧紧扭矩: 19.6~26.5Nm

(内六角: 12mm)



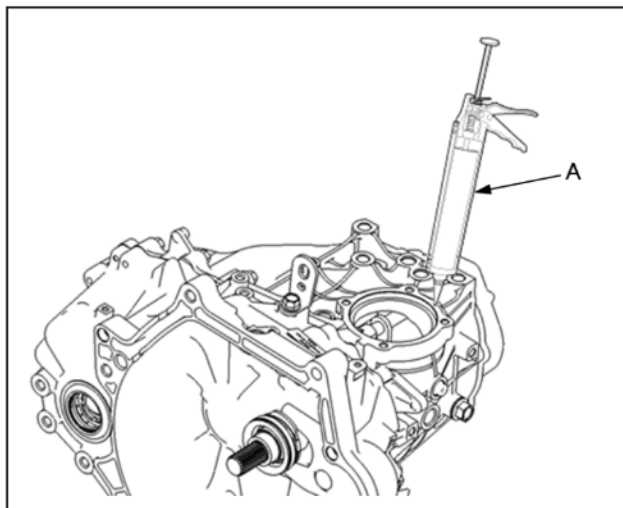
32) Place the shift lugs (A,B,C,D) as neutral position to install the control shaft complete.

将换挡杆(A,B,C,D)置于中立位置，并安装换挡总成



33) Clean the mating surface of transmission case and spread a fluid gasket (A).

清洗变速器壳体啮合面并涂密封胶(A)



34) Install control shaft complete(A) with 4-seal bolts(B) after spreading thread locker on the screw.

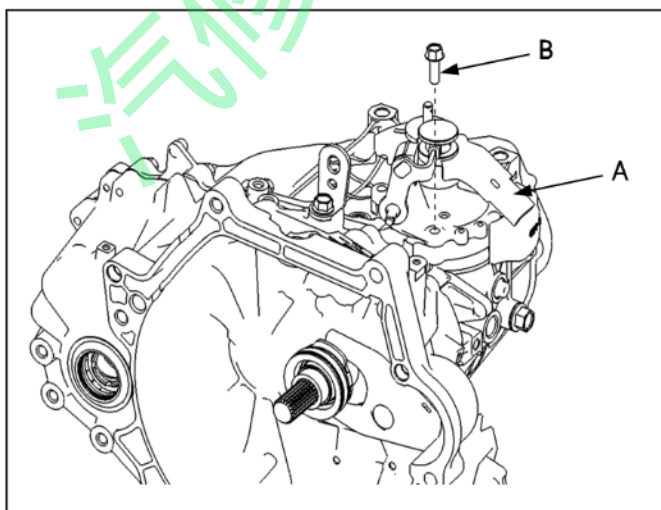
Tightening torque : 19.6 ~ 26.5Nm

(Hex socket : 12mm)

在螺钉上展开螺纹锁紧器后，在用 4 个密封螺栓(B)安装换挡总成(A)

拧紧扭矩 : 19.6 ~ 26.5Nm

(内六角 : 12mm)



35) Install 3-poppet spring assembly (A) after spreading thread locker on the screw.

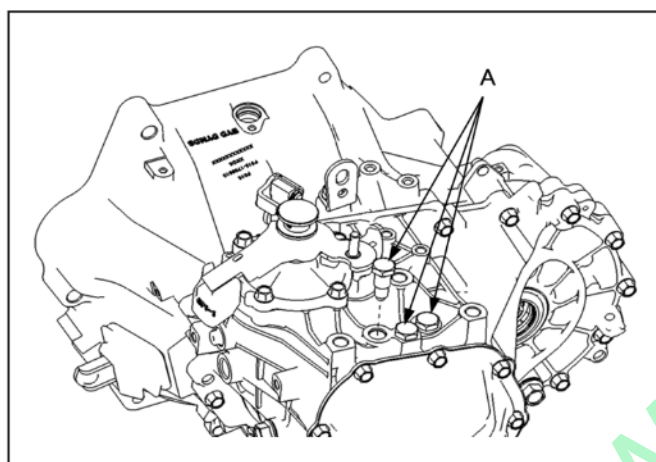
Tightening torque : 39.2 ~ 58.9Nm

(Hex socket : 19mm)

在螺旋桨上拧紧后安装 3 个弹簧组件(A)。

拧紧扭矩: 39.2~58.9Nm

(内六角 : 19mm)



36) Install back up lamp switch (A) after spreading thread locker on the screw.

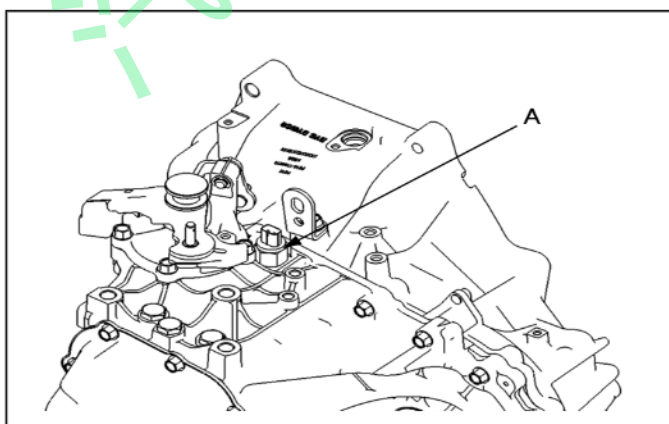
Tightening torque : 39.2 ~ 58.9Nm

(Hex socket : 24mm)

在螺钉上展开螺纹锁紧器后，插入倒档开关(A)

拧紧扭矩: 39.2~58.9Nm

(内六角: 24mm)



37) Install neutral switch (A) after spreading thread locker on the screw.

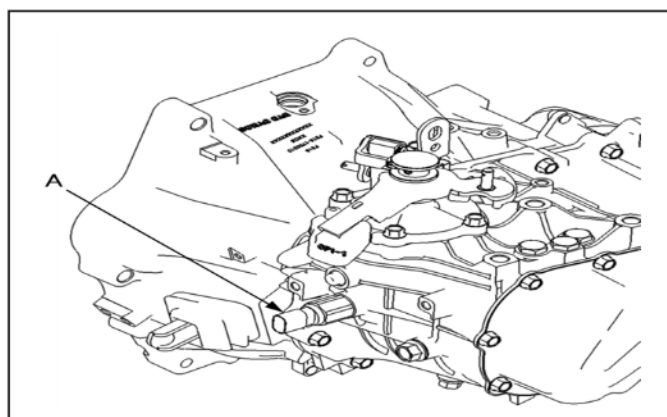
Tightening torque : 39.2 ~ 58.9Nm

(Hex socket : 25mm)

在螺钉上展开螺纹锁紧器后，插入空档开关(A)

拧紧扭矩：39.2~58.9Nm

(内六角：25mm)



38) Install cable bracket (A) with 3-seal bolts (B).

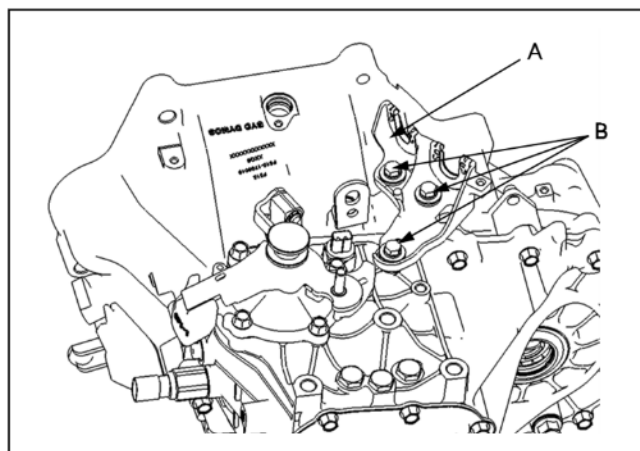
Tightening torque : 14.7 ~ 21.6Nm

(Hex socket : 12mm)

用 3 个密封螺栓(B) 安装线束支架(A)

拧紧扭矩：14.7~21.6Nm

(内六角：12mm)

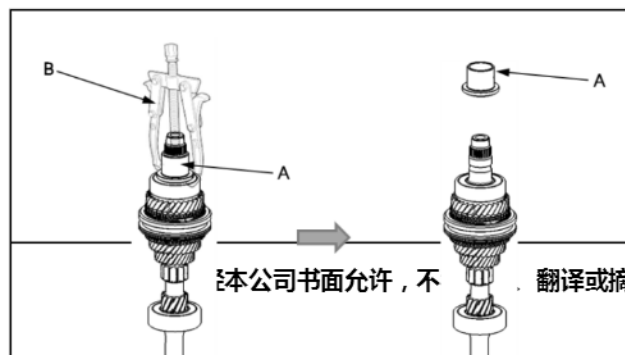


六、输入轴分总成的拆装

拆卸步骤

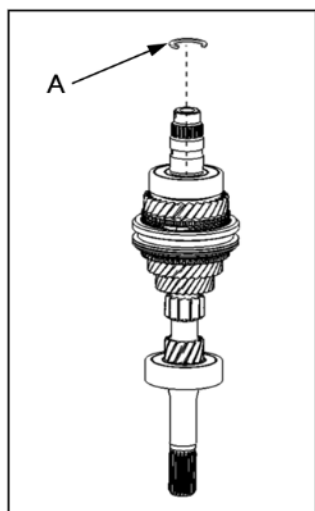
1) Remove the 5th sleeve (A) by using a gear puller (B).

使用扳子(B)移除 5 档齿套(A)



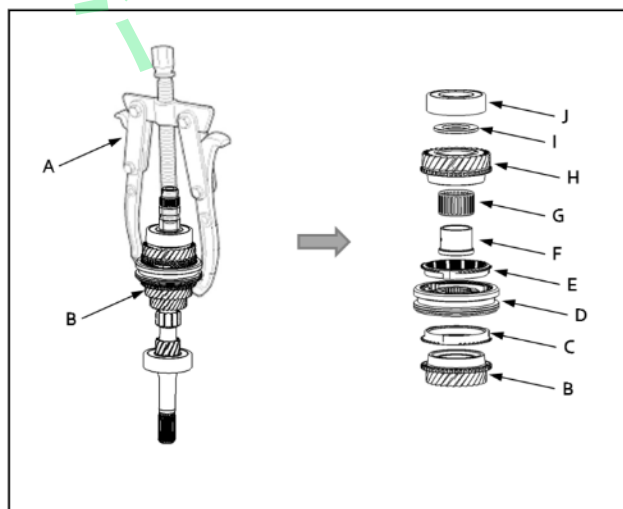
2) Remove the snap ring (A) by using plier.

使用卡簧钳 y 移除卡圈(A)



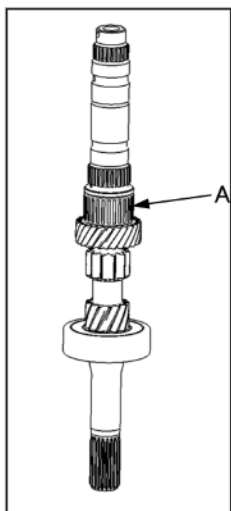
3) After installing a gear puller (A) to 3rd gear (B), disassemble the synchronizer ring (C) – 3&4th hub assembly (D) – synchronizer ring (E) – 4th sleeve (F) – needle bearing (G) – 4th gear (H) – thrust washer (I) – ball bearing (J) at once.

齿轮拆解器(A)装到3档齿轮(B)位置时, 移除同步器环(C) – 3&4档齿毂套件(D) – 同步器环(E) – 4档齿套(F) – 滚针轴承(G) – 4档齿轮(H) – 弹簧垫圈(I) – 滚珠轴承(J)



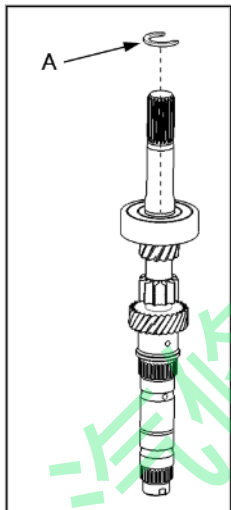
4) Remove the needle bearing (A).

移除滚珠轴承(A)



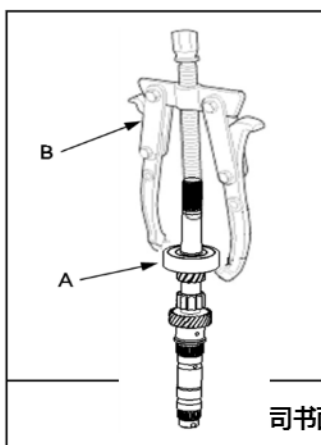
5) Remove the snap ring (A) by using plier.

使用卡簧钳移除卡圈(A)



6) Remove the ball bearing (A) by using a gear puller (B).

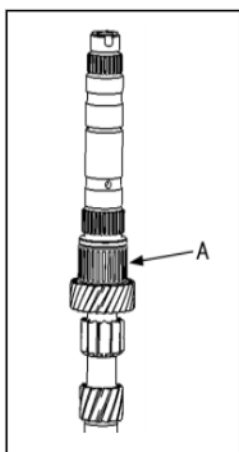
使用齿轮拆解器移除滚珠轴承(A)



安装步骤

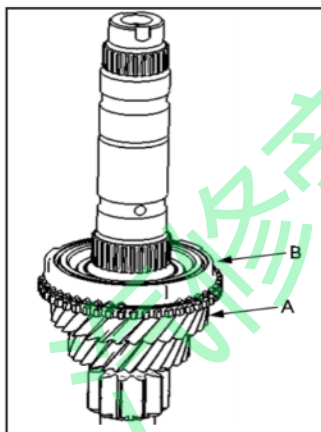
- 1) Insert the needle bearing (A).

插入滚针轴承(A)



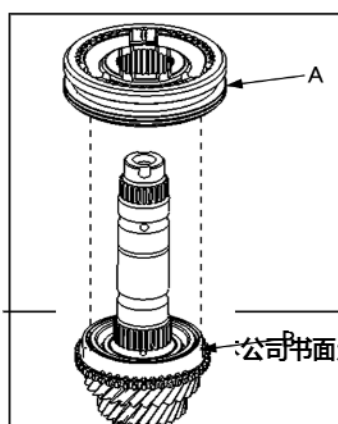
- 2) Insert the 3rd gear (A) and the synchronizer ring (B).

插入 3 档齿轮(A)和同步器环(B)



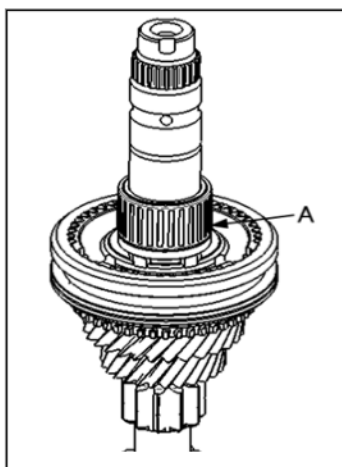
- 2) Press 3&4th hub assembly (A) after aligning boss of synchronizer ring (B) with slot of 3&4th hub.

校正同步器环并压入 3&4 档齿毂齿套(A)



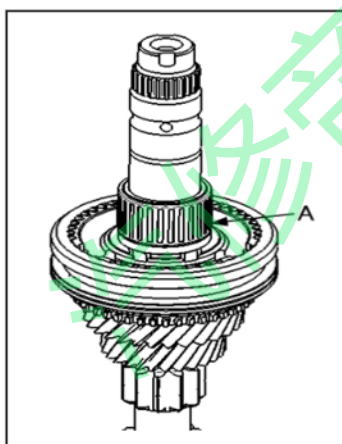
4) Press 4th gear sleeve (A).

压入 4 档齿套(A)



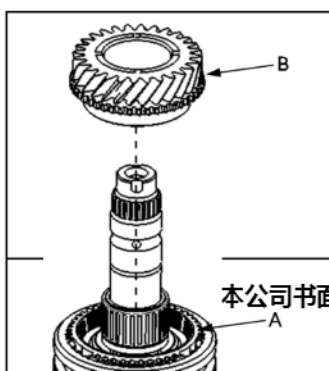
5) Insert 4th needle bearing (A).

插入 4 档滚针轴承(A)



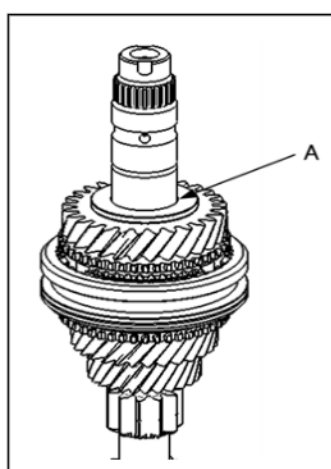
6) Insert synchronizer ring (A) and 4th gear (B).

插入同步器环(A)和 4 档齿轮(B)



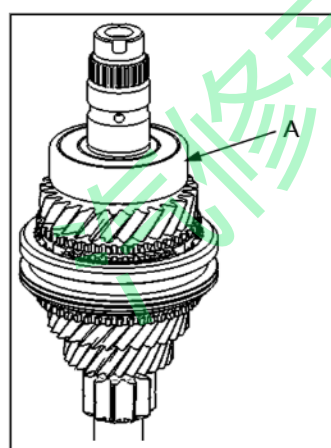
7) Insert thrust washer (A).

插入弹簧垫片(A)



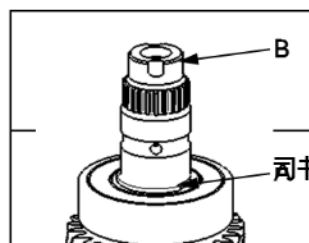
8) Press the ball bearing (A).

压入滚珠轴承(A)



9) Select the snap ring(A) to adjust the bearing clearance to 0 ~ 0.05mm and insert snap ring(A) in the slot of input shaft(B) by using snap ring plier.

选择同步器环(A)去调整轴承的间隙从 0~0.05mm, 并且用同步器拉马在输入轴(B)槽处插入同步器环(A)

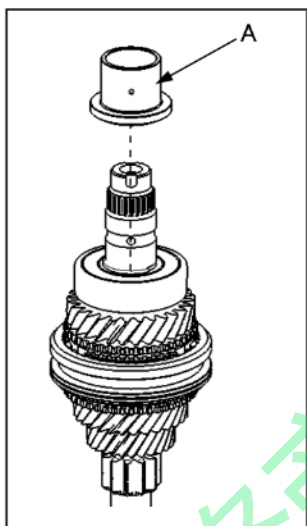


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Part number	Thickness[mm]	I.D Marks
43279-32188	⁴¹ 1.88	Green
43279-32192	1.92	Blue
43279-32196	1.96	Red

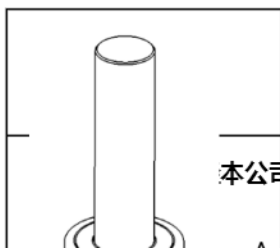
10) Press the 5th gear sleeve (A).

压入 5 档齿套(A)



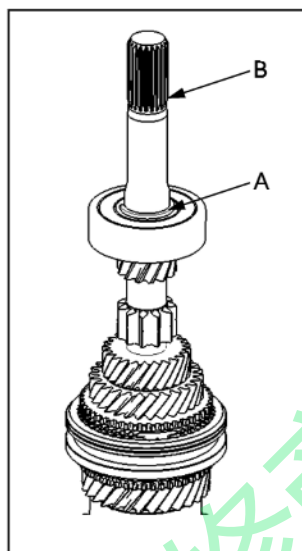
11) Press the ball bearing (A).

压入滚珠轴承(A)



12) Select the snap ring(A) to adjust the bearing clearance to 0 ~ 0.05mm and insert snap ring(A) in the slot of input shaft(B) by using plier.

选择卡环(A)将轴承间隙调整到 0~0.05mm，并用钳子在输入轴(B)的槽内插入卡环(A)



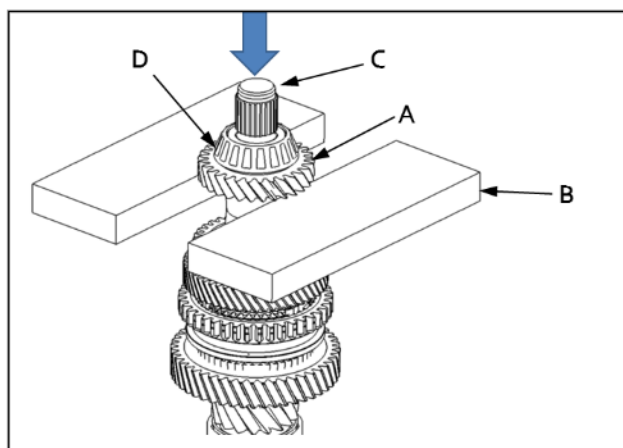
Part number	Thickness[mm]	I.D Marks
43224-26250	2.25	GREEN
43224-26280	2.28	BLUE
43224-26310	2.31	RED
43224-26340	2.34	- (NON)
43224-26370	2.37	BLACK
43224-26400	2.40	YELLOW
43224-26430	2.43	WHITE

七、输出轴分总成的拆装

拆卸步骤

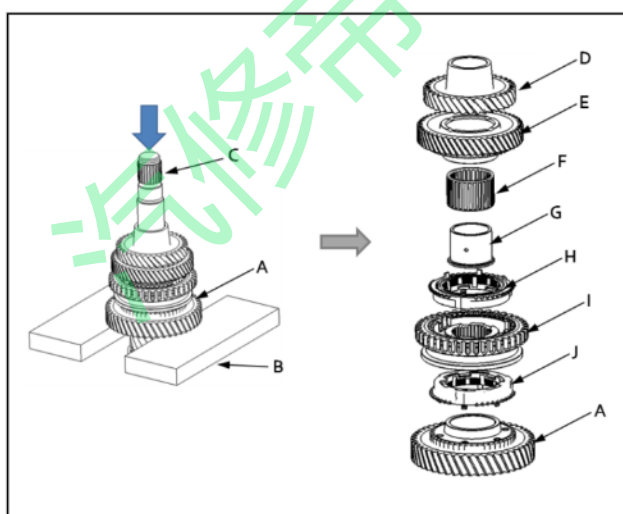
- 1) Place the 4th output gear (A) on the block (B) and press the shaft(C) with the press machine to remove the bearing(D) and the 4th output gear(A).

在工装(B)上置于 4 档输出齿轮(A)之下，用压力机压入轴(C)移除轴承(D)及 4 档输出齿轮(A)



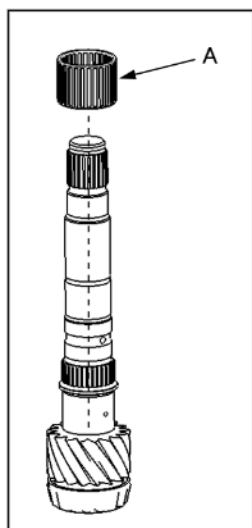
- 2) Place the 1st output gear(A) on the block(B) and press the shaft(C) with the press machine to remove the 3rd output gear(D) - 2nd output gear(E) - 2nd needle bearing(F) - 2nd gear sleeve(G) - synchronizer ring(H) - 1&2nd hub assembly(I) - synchronizer ring(J).

在工装(B)上置于 1 档输出齿轮(A)之下，用压力机压入轴(C)移除 3 档输出齿轮(D) - 2 档输出齿轮(E) - 2 档滚针轴承(F) - 2 档齿套(G) - 同步器环(H) - 1&2 档齿毂套件(I) - 同步器环(J)



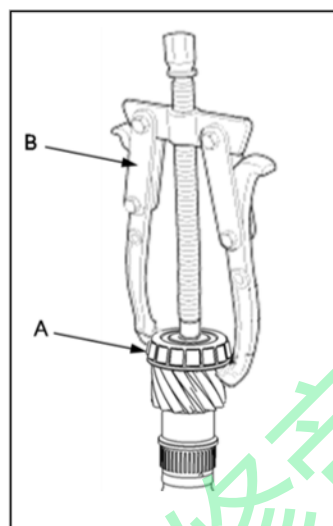
- 3) Remove the needle bearing (A).

移除滚针轴承(A)



4) Remove the taper roller bearing (A) by using a gear puller (B).

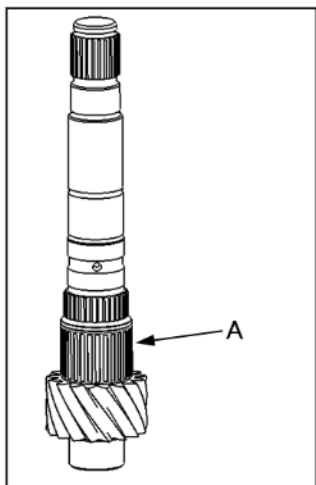
用齿轮分离器(B)移除圆锥滚子轴承(A)



安装步骤

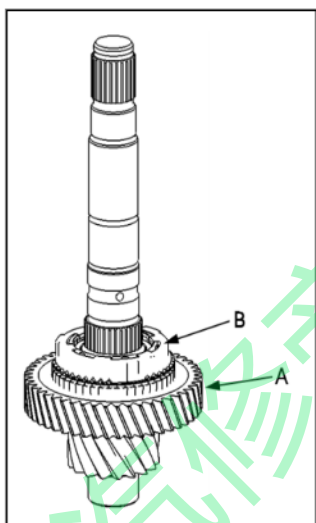
1) Insert the needle bearing (A).

插入滚针轴承(A)



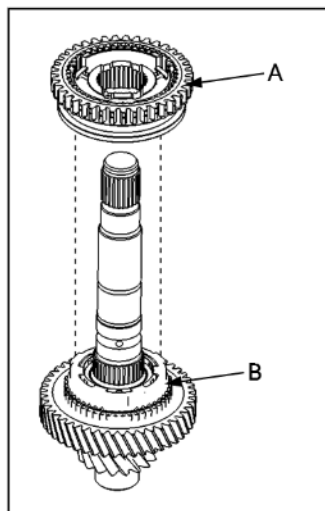
2) Insert the 1st gear (A) and the synchronizer ring (B).

插入 1 档齿轮(A)和同步器环(B)



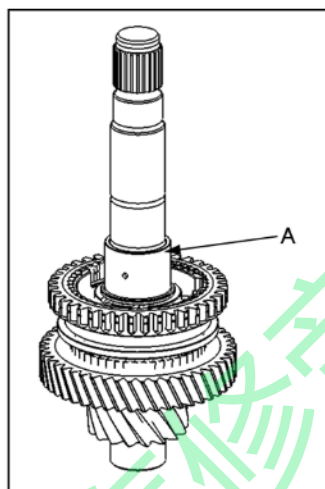
3) Press 1&2nd hub assembly (A) after aligning boss of synchronizer ring(B) with slot of 1&2nd hub.

校正同步器环(B)之后，在 1&2 档齿毂槽处插入 1&2 档齿毂组件(A)



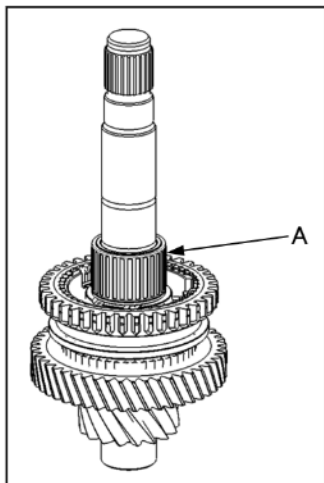
4) Press 2nd gear sleeve (A).

压入 2 档齿套(A)



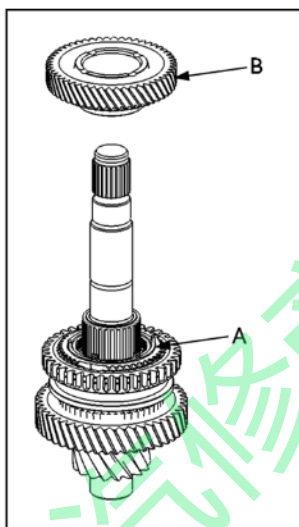
5) Insert 2nd needle bearing (A)

插入 2 滚针轴承 (A)



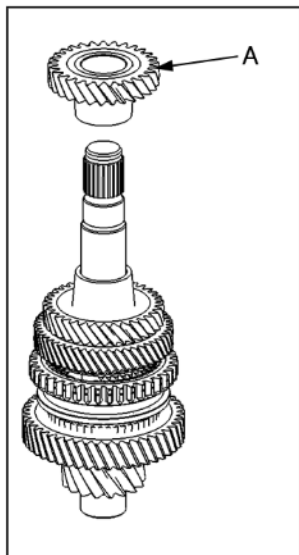
6) Insert synchronizer ring (A) and 2nd gear (B).

插入同步器环(A)和 2 档齿轮(B)



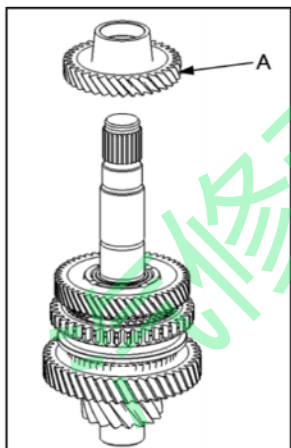
7) Press 3rd gear(A) after preheating it below maximum 150°C

在 150°C 下预热，后压入 3 档齿轮(A)



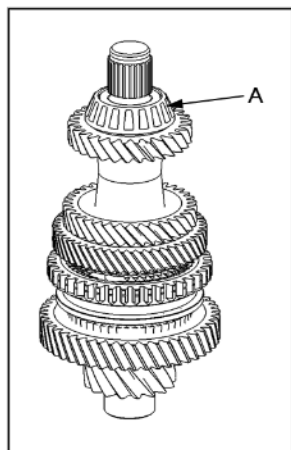
8) Press 4th gear(A) after preheating it below maximum 150°C

将 4 档齿轮(A)在最大温度 150°C 以下预热



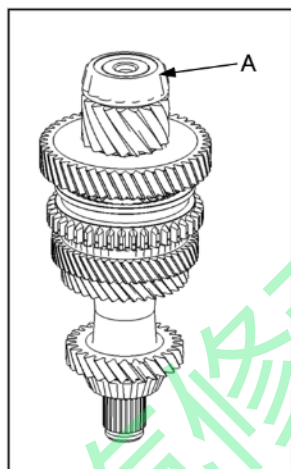
9) Press the taper roller bearing (A).

压入圆锥滚针轴承(A)



10) Press the taper roller bearing (A).

压入圆锥滚针轴承(A)

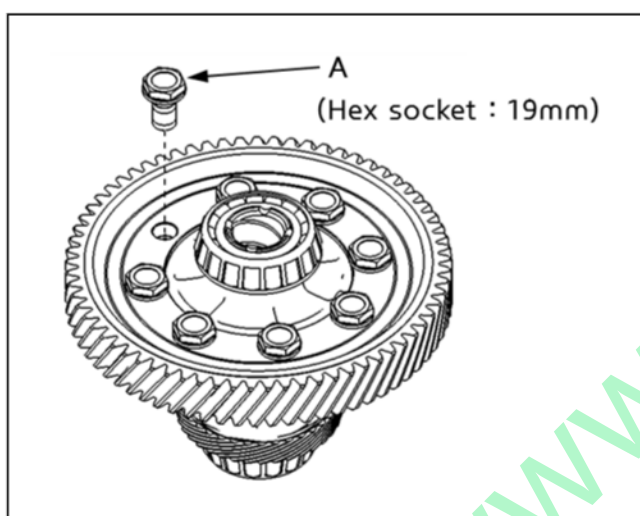


八、差速器分总成的拆装

拆卸步骤

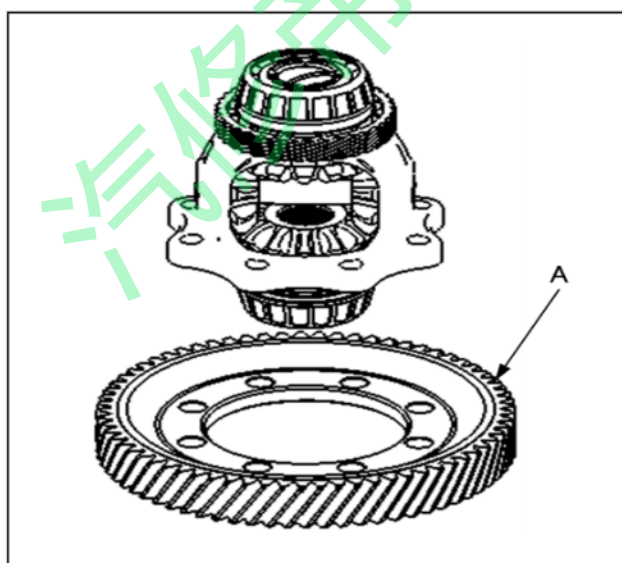
- 1) Remove 8-seal bolts (A) of the differential drive gear.

拆下差速传动齿轮的 8 个密封螺栓(A)



- 2) Disassemble the differential drive gear (A) from the differential case by using a rubber hammer or a gear puller.

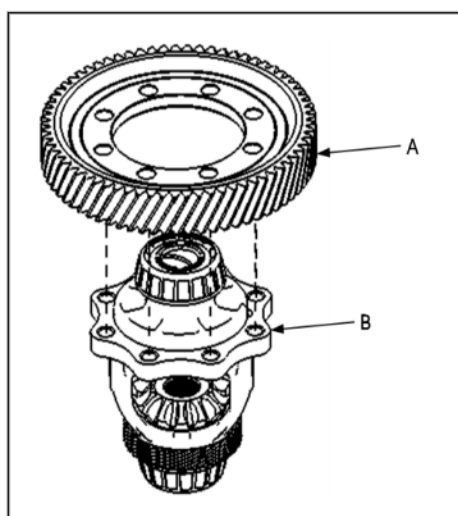
通过使用橡胶锤或齿轮拉拔器从差速器壳体拆卸差动传动齿轮(A)。



安装步骤

- 1) Insert differential drive gear (A) into the differential case (B).

将变速器驱动齿轮(A)插入变速器壳体(B)中。

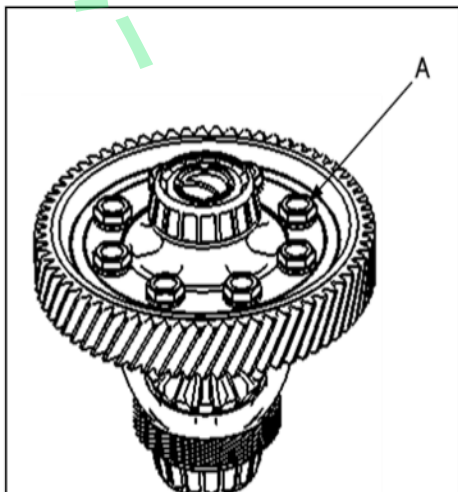


- 2) Install 8-seal bolts (A) of the differential drive gear.

Tightening torque : 127.5 ~ 137.3Nm (Hex socket : 19mm)

安装差速驱动齿轮的 8 个密封螺栓(A)。

拧紧扭矩：127.5 ~ 137.3Nm(内六角：19mm)



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